# PORTFOLIO CONSTRUCTION WITH BLACK-LITTERMAN MODEL AND FAMA-FRENCH FIVE FACTORS

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**ABSTRACT**

Optimizing trading strategies in the face of market efficiency and time series volatility poses significant challenges. This paper reports the methodology and findings for constructing an optimally diversified, factor-bearing portfolio by analyzing cross-sector ETFs using the Black-Litterman model, Fama-French factors, correlation analysis and portfolio optimization techniques. Investor views, informed by four global macroeconomic factors and the New Combined Return Vector, serve as priors in the Black-Litterman model, dynamically adjusting portfolio weights. We employ optimization techniques including mean-variance minimization, Maximum Sharpe Ratio, higher-order moments, and Minimum Tracking Error against the SPY benchmark to create four optimal portfolios, driven by investor views and market data confidence levels. Risk aversion is evaluated through three distinct investor profiles—Trustee, Market, and Kelly Investor—each reflecting varying degrees of risk tolerance. Incorporating the Kelly criterion into our methodology further refines the portfolio by favoring assets with long-term growth potential. Systematic backtesting regresses portfolio performance against market excess returns, offering insights into the impact of these factors on the factor-based portfolio.

***Keywords*** Portfolio Construction · Optimization · Factors Investing

## 1 Introduction

In this final project, as part of the academic requirement for the Certificate in Quantitative Finance (CQF) program, we study the construction of a factor-bearing diversified optimal portfolio that integrates confidence level-based investor views into the Black-Litterman model. The Black-Litterman model combines market equilibrium returns with investor views through a Bayesian framework, adjusting expected returns while accounting for the uncertainty of the views (see Figure 1). These calculations are detailed in this report, blending market equilibrium returns with subjective views to produce optimal weight allocations using adjusted expected returns.

The investor views are informed through both a macroeconomic analysis and a rationale given for the current market views, which adjusts the portfolio's weights based on observed macroeconomic trends historically and into the near future. While the model primarily adjusts the expected returns, it also accounts for the uncertainty of these views, which impacts the portfolio's risk profile.

Our data collection involves twenty-two (22) U.S. equities, sector-specific ETFs, and notionally uncorrelated assets such as commodities, emerging and developed markets, VIX, bonds, energy, and real estate (see Table 1). Out of the twenty-two (22) assets, we compute initial weights based on 18 Factors, 1 Benchmark (ACWI), and 1 Emerging Markets (EEM) to make up a total of twenty (20) assets, excluding SPY benchmark and RFR. Using the correlation analysis of these assets, we narrow the Portfolio Choice down to eight (8) 'optimal assets' using a heuristically determined correlation threshold. This portfolio selection approach follows *Option A* as outlined in the Project QA.

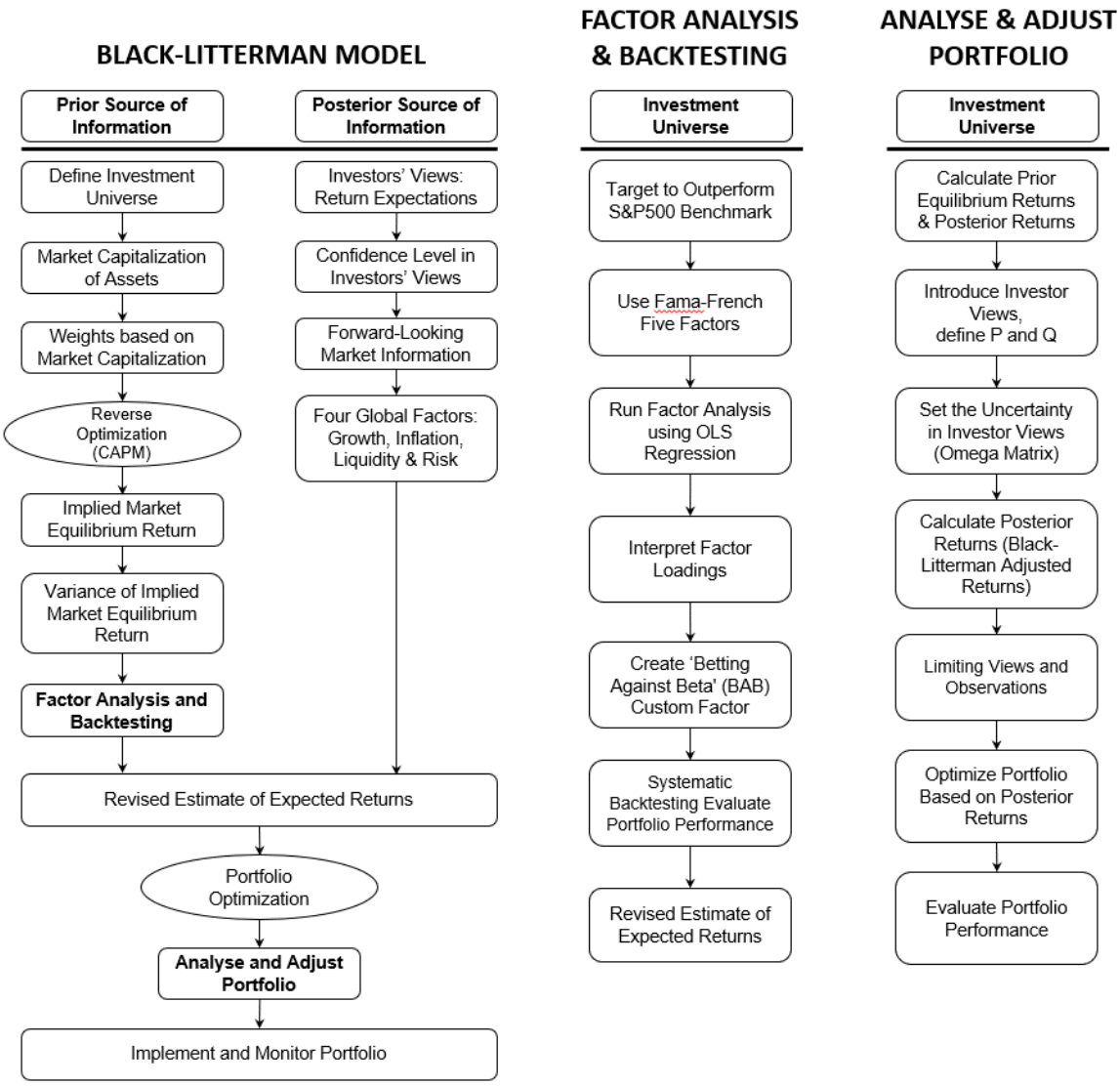



Figure 1: Flowchart on Black-Litterman Model Process for Portfolio Construction and Optimization

The risk-free rate (RFR) is obtained using the time series on the 13-week Treasury bill (^IRX) via *yahooquery* and is used to adjust returns for both the CAPM and Fama-French factors. When implementing the Fama-French five-factor model, the RFR is further derived from the 'RF' column in the Fama-French CSV dataset. This accurate risk-free rate adjustment is critical for analyzing market exposures and calculating excess returns in regression models.

We implement the Fama-French five-factor model, which captures exposures to five key risk factors—market (Mkt-RF), size (SMB), value (HML), profitability (RMW), and investment (CMA). Using a regression-based technique, we estimate factor loadings and adjust the portfolio's asset weights accordingly. To enhance risk-adjusted returns, we tilt the portfolio toward favorable exposures, such as small-cap value stocks, and introduce a custom factor, Betting Against Beta (BAB), to differentiate between high- and low-beta assets, further capturing risk-adjusted excess returns.

To ensure robustness, systematic backtesting is conducted to 1) evaluate the dynamic behavior of each of the twenty (20) ETF's returns, optimized portfolios' returns, and BAB returns, 2) optimize the portfolio's performance, and 3) analyze rolling beta and alpha over time relative to the SPY benchmark. This process helped us identify time-varying factor exposures, which informed strategic allocation adjustments.

We calculate the CAPM excess returns relative to ACWI (a proxy for the global market), and we evaluate extreme daily returns (over 2.5%) to detect anomalies. We further calculate historical expected returns, adjust for the risk-free rate, and compare these results to the implied market equilibrium returns (Π) derived from the Black-Litterman framework. This equilibrium return calculation is grounded in market indices like MSCI and S&P Dow Jones and serves as a foundation for integrating investor views.

The Black-Litterman model is applied to a diverse portfolio of ETFs, incorporating investor views on equities and alternative assets such as commodities, bonds, and real estate. We adjust these views using confidence levels to reflect market sentiment and combine these insights with macroeconomic analysis to fine-tune the portfolio's allocations. The implied equilibrium returns (Π) are adjusted using these subjective views, and quadratic programming is employed to optimize the final portfolio weights.

Four optimization techniques are used to construct the portfolio – 1) Mean-Variance Optimization: minimizes risk for a given return; 2) Maximum Sharpe Ratio: optimizes risk-adjusted returns; 3) Higher-Order Moments: focuses on minimizing coskewness and maximizing cokurtosis; and 4) Minimum Tracking Error: ensures minimal deviation from the SPY benchmark. These optimization methods are tailored to three levels of risk aversion—Trustee, Market, and Kelly Investor—to ensure the portfolio meets diverse investment goals and is resilient across varying risk profiles. The portfolio optimization process is handled using Python's scipy.optimize.minimize function to solve for the optimal weights efficiently and accurately.

In addition, we introduce the Kelly Criterion into our methodology, which maximizes long-term wealth growth by favoring assets with better growth potential. This criterion complements our factor-based approach and further refines the portfolio's allocations to favor long-term returns.

## 2 Report Structure

This report outlines our approach to portfolio construction, enhancing traditional methods with advanced models that consider factor-based analysis and risk sensitivities. We integrate the Black-Litterman model and the Fama-French five-factor model to refine factor selection and portfolio optimization. The report follows a structured methodology, leveraging numerical techniques, correlation analysis, and various portfolio optimization strategies. Section 3 reviews foundational work on portfolio optimization, technical issues in data collection, and detailed the Python package imports, and initial setups are detailed in the programming environment for financial data collection, portfolio construction and financial modeling. A table shows the mapping of mathematical symbols to Python symbols used in the Project Code. Section 4 describes our methodology, showing tables and figures on the Black Litterman process and the numerical techniques we have coded and utilized, and goes into more details in the sub-sections. The sub-sections explained the Black-Litterman process with mathematical formulas and a walkthrough of how investor views and market equilibrium returns are incorporated to adjust portfolio weights. We outline on trial-and-error in using various data sources, results, correlation analysis, investor views, different investor profiles, showcasing how portfolio weights adjust under different risk aversion scenarios, along with the impact of integrating Fama-French factors, integrating the Kelly Criterion into the portfolio strategy, and results from backtesting and portfolio performance. Section 5 offers a discussion of the future work and potential improvements in risk-sensitive asset management, the use of LSTM to predict future expected returns, and the possibility of using deep learning to generate investor views. Section 6 provides the conclusion linked to the findings, current limitations, and recommendations for future research.

## 3 Related Works, Issues, Technical Framework & Python Libraries

In this section, we provide a comprehensive overview of the Python implementation of numerical techniques for portfolio construction using the Black-Litterman model and the Fama-French Five Factors (refer to Table 3). Under

Related Works section, we discuss about the Black-Litterman model and the Fama-French five-factor model, with references to prior academic papers and industry practices. In technical issues, we detailed the problems encountered during the data collection stage and how a workaround was implemented.

### 3.1 Related Works

The Black-Litterman model, introduced by Fischer Black and Robert Litterman in 1992, addressed the practical challenges associated with portfolio optimization in the Markowitz mean-variance framework. The Black-Litterman model allows for the integration of investor views into market equilibrium returns, resolving the instability and sensitivity issues often encountered in mean-variance optimization, particularly when small changes in input data occur (Black and Litterman, 1992). Later work by Idzorek (2007) provided a systematic approach to assigning confidence levels to investor views, further enhancing the model's practical application in portfolio construction.

Parallel to these developments, the Fama-French Three-Factor Model was introduced by Eugene Fama and Kenneth French in 1993, extending the Capital Asset Pricing Model (CAPM) by adding three factors: size (SMB), value (HML), and market risk premium (Mkt-RF) (Fama and French, 1993). These three factors became fundamental for explaining excess returns beyond CAPM, especially in portfolio construction. In 2015, Fama and French expanded their model to five factors, namely profitability (RMW) and investment (CMA), to further capture drivers of stock returns (Fama and French, 2015). Recent research by Lin (2020) built upon this model by combining it with multi-factor construction approaches and emphasizing how factor loadings can be used to tilt portfolios toward specific risk premiums. On the other hand, Li and Shim (2020) identify six factors (value, low beta, profitability, investment, momentum, and size) that contribute diversification benefits to optimal portfolio construction.

The foundation of modern portfolio theory was laid by Harry Markowitz (1952) who introduced mean-variance optimization as a method for constructing portfolios that maximize expected return for a given level of risk. However, practical implementation often faces challenges due to the instability of covariance matrix estimates and the difficulty in forecasting returns. In response, factor investing has emerged as a prominent strategy in asset management. Contributions from Ang, Goetzmann, and Schaefer (2009) argue that factors represent risk premia that investors can systematically capture through thoughtful portfolio construction.

### 3.2 Issues with Yahoo! Finance and Nasdaq.com website

While attempting to download market capitalization data using the *yfinance* package, we encountered the error: "Exception: yfinance failed to decrypt Yahoo data response." This issue, documented in the GitHub repository here, remains unresolved at the time of writing. As a result, we have opted to use the alternative Python package, *yahooquery*, to retrieve daily and monthly OHLCV financial data reliably.

Additionally, any efforts to extract market capitalization data from Nasdaq.com resulted in an IP ban due to their implementation of anti-scraping mechanisms. Despite trying Selenium webdriver, relevant market cap data remains hidden in the JSON response, making it inaccessible through standard scraping methods. Furthermore, many websites like Nasdaq.com, track the User-Agent information in requests; frequent requests with the same User-Agent lead to IP bans. To circumvent these issues, we opted to use Kavout.com for market capitalization data. In cases of connection timeouts or retry errors with Kavout.com, we discovered that manually visiting the first URL (e.g., 'https://www.kavout.com/etfs/spy/vanguard-growth-etf') to "unlock" access is necessary:

```
[**********100%%**********]  1 of 1 completed
https://www.kavout.com/etfs/spy/vanguard-growth-etf
```

Once this step is completed, rerunning the script allows access to market cap data without further disruptions.

Table 1: Overview of Factors, Benchmark, and Risk-Free Rate (22 Time Series: 18 Factors, 2 Benchmarks, 1 Emerging Markets, and 1 Risk-Free Rate)

| No. | Type | Symbol | Description | Factor | Filter^ |
|---|---|---|---|---|---|
| **1** | US Equity | SPY | Represents US S&P 500 Index. | Benchmark | x |
| **2** | US Equity | VUG | Focuses on growth stocks in the U.S. | US Growth | o |
| **3** | US Equity | VTV | Represents value stocks in the U.S. | US Value | x |
| **4** | US Equity | SPHD | Targets U.S. stocks with high dividend yields and low volatility. | US Dividend | o |
| **5** | US Equity | VB | Focuses on small-cap U.S. stocks. | US Small Cap | x |
| **6** | US Equity | MTUM | Targets U.S. stocks demonstrating higher price momentum. | US Momentum | o |
| **7** | Global Equity | ACWI | Represents global stocks. | Global Benchmark | o |
| **8** | Emerging Markets | EEM | Tracks MSCI Emerging Markets Index. | EM Benchmark | x |
| **9** | Developed (ex. US) | VEA | Covers developed markets, excluding the US. | Developed Ex-US | x |
| **10** | Developed (ex. US) | EFA | Exposure to Europe, Australasia, and Far East. | EAFE | x |
| **11** | Financial | XLF | Focus on the financial sector. | Financial Sector | x |
| **12** | Technology | XLK | Exposure to technology sector. | Technology Sector | o |
| **13** | Energy | XLE | Focus on energy companies. | Energy Sector | o |
| **14** | Oil & Gas | IEO | Focuses on U.S. companies in the Oil & Gas sector. | Oil & Gas Sector | o |
| **15** | US Treasury | SCHO | Exposure to short-term U.S. Treasury securities. | ST US Treasury | x |
| **16** | US Bonds | TLT | Tracks long-term U.S. Treasury bonds. | LT US Treasury | x |
| **17** | Commodities | GLD | Exposure to gold. | Gold | o |
| **18** | Commodities | USO | Tracks the performance of crude oil. | Crude Oil | x |
| **19** | Real Estate | IYR | Exposure to U.S. real estate sector. | US Real Estate | x |
| **20** | Real Estate | VNQ | Focuses on U.S. real estate investment trusts (REITs). | US REITs | x |
| **21** | Volatility | VIXY | Provides exposure to short-term volatility based on VIX. | Short-Term Volatility | x |
| **22** | Risk Free Rate | RFR | The 13-week U.S. Treasury Bill | US Treasury Rate | x |

^ Filtering on selected assets used in portfolio choice restricted to less than 10 represented by symbol ‘o’.

Table 2: Mathematical symbols to Python symbols:

| Python Symbol | Math Symbol | Description |
|---|---|---|
| rfr, risk_free_rate | $r$ | Risk free rate |
| risk_aversion | $\lambda$ | Risk aversion coefficient |
| C | $C$ | Assets covariance matrix |
| R, returns | $R$ | Assets returns |
| W, W_a | $W$ | Assets weights |
| asset_returns | $\mu$ | Portfolio historical return |
| Pi | $\Pi$ | Portfolio equilibrium excess returns |
| adjusted_returns | $\mu$ | Posterior expected returns (adjusted returns) in the Black-Litterman model |
| adjusted_covariance | $\Sigma$ | Covariance matrix of asset returns, representing asset risk relationships |
| sigma | $\sigma$ | Volatility or standard deviation |
| tau | $\tau$ | Scaling factor for investor confidence in the Black-Litterman model |
| omega | $\Omega$ | Uncertainty matrix of investor views in the Black-Litterman model |

### 3.4 Technical Framework

This section outlines how we apply key portfolio optimization techniques, model-building processes, and correlation analysis within Python, leveraging robust industry-grade libraries to implement these advanced methods. The methodology includes several core processes, such as correlation analysis, model building, covariance matrix calculations, and performance evaluation, which are critical for developing a systematic and optimized portfolio. Python's flexibility allows us to implement these techniques effectively, starting from data collection to the application of advanced portfolio optimization routines like quadratic programming.

To support the execution of these numerical techniques, we use a set of powerful Python libraries commonly adopted in industry-level financial analytics. All necessary libraries—such as pandas for data manipulation, numpy for numerical computations, matplotlib and seaborn for data visualization, and tabulate for organized output presentation—are imported at the beginning of the code. These libraries ensure efficient handling of large datasets, accurate numerical calculations, and seamless integration of visualization components, which are essential for interpreting the results of the portfolio construction.

In Table 3, we present a detailed breakdown of the numerical techniques implemented, highlighting the Python routines used to model the Black-Litterman and Fama-French five factor frameworks. These techniques include data preparation, regression-based factor analysis, the calculation of implied returns, covariance matrices, and the optimization of portfolio weights using quadratic programming routines to ensure efficient and accurate portfolio allocation. Additionally, the integration of investor views and the adjustment of expected returns using the Black-Litterman model are detailed, demonstrating how Python enables the practical application of these advanced financial models.

Table 3: Python implementation of numerical techniques from first principles for Portfolio Construction using the Black-Litterman model and Fama-French Five Factors

| No | Type | Numerical Technique | Python Implementation | Explanation |
|---|---|---|---|---|
| **1** | Data Collection | Yahoo Finance API for ETFs financial data | yahooquery (for ETF prices), pandas (to load CSV and dataframe) | Uses the Yahoo Finance API to fetch historical ETF prices, risk-free rate, and the Fama-French dataset for factors such as SMB, HML, RMW, and CMA. |
| **2** | Risk-Free Rate (RFR) - Daily | Data Fetching from Yahoo Finance | tnx = yf.Ticker('^IRX')<br># Get historical market data<br>hist = tnx.history(period="5d")<br>rfr = float(hist['Close'].iloc[-1]) / 100 | Retrieves the latest 13-week Treasury Bill (^IRX) yield using yahooquery. It fetches the historical data for the past 5 days, extracts the most recent 'Close' value and converts it from a percentage to a decimal by dividing it by 100. |
| **3** | Risk-Free Rate (RFR) - Monthly | Data Fetching from Fama-French Dataset[1] | factors = pd.read_csv('data/F-F_Research_Data_5_Factors_2x3_monthly.csv', index_col=0, parse_dates=True)<br>rfr = factors['RF'] | Fama-French CSV dataset contains monthly risk-free rate. Uses monthly RFR to calculate excess returns for each ETF, then regressed against the Fama-French factors to determine the ETF's sensitivity to systematic risk. |
| **4** | Expected Returns (Black-Litterman) | Implied Returns from CAPM | Pi = risk_aversion * np.dot(cov_matrix, market_weights) | Derives market equilibrium returns using CAPM, where $\Pi = \lambda \Sigma W_a$ , with market weights ($W_a$) and covariance matrix ($\Sigma$) obtained from historical returns. Uses risk_aversion = 3.07. |
| **5** | Investor Views Representation (Black-Litterman) | Views Matrix Q, Link Matrix P | Q, P = create_views_and_link_matrix (etf_names, investor_views) | This function creates the P matrix to represent relative ETF performance expectations and the Q vector for expected excess returns, based on investor views. |

[1] Fama-French Data retrieved from: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

| No | Type | Numerical Technique | Python Implementation | Explanation |
|---|---|---|---|---|
| **6** | Relative Investor Views Construction (Black-Litterman) | Investor Views | investor_views = [ ('GLD', '>', 'MTUM', 0.15), ('XLK', '>', 'XLE', 0.08), ('SPHD', '>', 'VUG', 0.06) ] | List of investor views, each tuple represents a belief that one ETF will outperform another by a specific margin. E.g. ('GLD', '>', 'MTUM', 0.15) means the investor expects GLD to outperform MTUM by 15%. |
| **7** | Covariance Matrix | Covariance (C) Calculation | C = assets_historical_returns_and _covariances(asset_returns) | This function calculates the general covariance matrix of the asset returns using historical data. |
| **8** | Covariance Matrix | Covariance (sigma_scaled) Calculation | tau = 0.1<br>sigma_scaled = tau * C | Scaled covariance matrix reflects the relationships between the returns of assets in the portfolio and is used in conjunction with the views matrix (P) and uncertainty matrix (omega) for calculating posterior returns in the Black-Litterman model. |
| **9** | Risk Aversion Coefficient λ | Empirical Value for U.S. Market (Idzorek, 2007) | risk_aversion = 3.07 | Sets the risk aversion coefficient to 3.07 based on empirical studies of the U.S. market. This adjusts the implied returns based on market risk aversion. |
| **10** | Black-Litterman Posterior Returns | Bayesian Updating for Posterior Returns | adjusted_returns = Pi + sigma_scaled @ P.T @ np.linalg.inv(P @ sigma_scaled @ P.T + omega) @ (Q - P @ Pi) | Uses the Black-Litterman formula: $\mu = \Pi + \Sigma P^T (P \Sigma P^T + \Omega)^{-1} (Q - P \Pi)$ to update expected returns based on the investor views, covariance matrix, and market equilibrium returns. |
| **11** | Covariance Matrix (Black-Litterman) | Posterior Covariance Update | adjusted_covariance = C + sigma_scaled - sigma_scaled @ P.T @ np.linalg.inv(P @ sigma_scaled @ P.T + omega) @ P @ sigma_scaled | Adjusts the covariance matrix to reflect the uncertainty in investor views, using the Black-Litterman posterior covariance formula. |
| **12** | Portfolio Weights | Initial Weights & Adjusted Weights | W = latest_caps / latest_caps.sum()<br><br>if W_a[ticker] > cap_threshold:<br>excess_weight = W_a[ticker] - cap_threshold | W represents the initial weighted using market captialization. W_a represents the adjusted weights based on ensuring no single ETF exceeds 10% of the portfolio. |
| **13** | Optimization (Mean-Variance) | Quadratic Programming | result = minimize(portfolio_variance, W_a, args=(C,), method='SLSQP', bounds=bounds, constraints=constraints) | Uses quadratic optimization to minimize portfolio variance while considering adjusted returns and the covariance matrix from the Black-Litterman model. |
| **14** | Optimization (Max Sharpe Ratio) | Max Sharpe Ratio via Quadratic Optimization | result_max_sharpe = minimize(neg_sharpe_ratio, initial_weights, args=(BL_return _vector, C, annual_rfr), method='SLSQP', bounds=bounds, constraints=constraints) | Maximizes the Sharpe ratio by quadratic optimizing the initial portfolio weights, where the Sharpe ratio is computed as $SR = \frac{r_p - r_f}{\sigma_p}$. |
| **15** | Optimization (Min. Tracking Error Minimization) | Quadratic Optimization of Min. Tracking Error Against Benchmark | result = minimize(calculate_tracking_ error, initial_weights, args=(portfolio_ returns, benchmark_returns), method='SLSQP', bounds=bounds, constraints=constraints) | Minimizes the tracking error by optimizing portfolio weights to track a benchmark (e.g., SPY) while minimizing deviations from the benchmark returns, with constraints like no short selling. |
| **16** | Optimization (Higher Order Moments - minimizing coskewness) | Quadratic Optimization of Higher Order Moments | result = minimize(lambda weights: - utility_function(weights, expected_returns, cov_matrix, coskew_matrix, cokurt_matrix, risk_aversion), initial_weights, method='SLSQP', bounds=bounds, constraints=constraints) | This optimization seeks to minimize the portfolio's coskewness, which reduces downside risk by controlling asymmetry in the distribution of returns. |
| **17** | Optimization (Higher Order Moments - maximizing cokurtosis) | Quadratic Optimization of Higher Order Moments | result = minimize(lambda weights: utility_function(weights, expected_returns, cov_matrix, coskew_matrix, cokurt_matrix, risk_aversion), initial_weights, method='SLSQP', bounds=bounds, constraints=constraints) | Maximizing cokurtosis helps reduce the likelihood of extreme negative outcomes (fat tails) in the portfolio's return distribution. |
| **18** | Factor Loadings (Fama-French) | OLS Regression for Factor Loadings | X = sm.add_constant(factors[['Mkt-RF', 'SMB', 'HML', 'RMW', 'CMA']]) y = returns[etf] - factors['RF'] model = sm.OLS(y, X).fit() | Uses ordinary least squares (OLS) regression to compute factor loadings of each asset on Fama-French five factors. |
| **19** | Factor-Based Portfolio Adjustment (Fama-French) | Threshold Filtering | filtered_loadings_df = factor_loadings_df.applymap( lambda x: x if (x >= pos_threshold or x <= - neg_threshold) else np.nan ) | Apply thresholds to filter significant factor loadings. pos_threshold = 0.010 and neg_threshold = 0.008. |

| No | Type | Numerical Technique | Python Implementation | Explanation |
|---|---|---|---|---|
| **20** | Factor-Based Portfolio Adjustment (Fama-French) | Dynamic Weight Adjustment | adjusted_weights = adjust_weights_based_on_factors(W_a, filtered_loadings_df) | Dynamically adjust portfolio weights based on the exposure of each ETF to these risk factors. |
| **21** | Investor View Uncertainty (Ω) | Diagonal Variance Matrix (Ω) | confidence_levels = [0.90, 0.70, 0.80] omega = np.diag([1 / c for c in confidence_levels]) | Represents the uncertainty in each investor view using a diagonal matrix, where confidence levels are provided for each view. |
| **22** | Black-Litterman Posterior Returns (BL_return_vector) | Bayesian Updating (Black-Litterman Model) | BL_return_vector = Pi + sigma_scaled.dot(P.T).dot(np.linalg.inv(P.dot(sigma_scaled).dot(P.T) + omega).dot(Q - P.dot(Pi))) | Represents the adjusted expected returns after incorporating investor views. |
| **23** | Optimal Portfolio Weights (Black-Litterman) | Quadratic Optimization Using Posterior Returns | result = minimize(neg_sharpe_ratio, initial_weights, args=(BL_return_vector, sigma_scaled, annual_rfr), method='SLSQP', bounds=bounds, constraints=constraints) | Solves the optimization problem for portfolio weights using the Black-Litterman adjusted returns and covariance matrix, with constraints like no short selling. |
| **24** | Factor Model (Five-Factor) | Calculation of Fama-French Five Factor Exposure | model = sm.OLS(y_aligned, X_aligned).fit() factor_loadings[etf] = model.params.drop('const').squeeze() | Regresses asset returns on the five Fama-French factors (Mkt-RF, SMB, HML, RMW, CMA) to compute exposures. For each ETF, adjust the weights based on the factor loadings, e.g., increase the weight if a factor needs to be maximized; decrease the weight if a factor needs to be minimized. |
| **25** | Backtesting (Factor Model) | Rolling Beta | window_data = asset_returns.iloc[i-rolling_window:i] X = sm.add_constant(spy_returns.iloc[i-rolling_window:i]) y = window_data[ticker] model = sm.OLS(y, X).fit() betas.append(model.params[1]) | Implements backtesting by calculating rolling betas and measures the sensitivity of the asset's return to the SPY market return. |
| **26** | Backtesting (Factor Model) | Factor Analysis | rolling_betas[ticker] = pd.Series(betas, index=asset_returns.index[rolling_window:]) ; rolling_alphas[ticker] = pd.Series(alphas, index=asset_returns.index[rolling_window:]) | The rolling regression calculates the alpha and beta values over time, which helps in understanding how each asset (ETF) is influenced by the market factor (SPY). |
| **27** | Compute Optimal Allocations (risk aversion levels) | Levels of risk aversion: Trustee / Market / Kelly Investor | risk_aversion_levels = {"Trustee": 3, "Market": 1, "Kelly": 0.5} | Allocates portfolio weights based on different levels of risk aversion. The "Trustee" is more conservative, while "Kelly" reflects an aggressive strategy based on the Kelly criterion. |
| **28** | Kelly Criterion | Maximizing long-term wealth | kelly_weights = inv_cov_matrix @ excess_returns; kelly_weights /= np.sum(np.abs(kelly_weights)) | Uses the Kelly criterion to determine optimal portfolio weights, aimed at maximizing the long-term growth of capital based on expected excess returns and the covariance of assets. |

# 4 Methodology

In this section, we describe our methodology in portfolio construction. Our goal is to maximize the expected log of the terminal wealth or expected rate of return using the utility function, a generalization of the Kelly[2], with an emphasis on accurately capturing the risk premium:

$$U_f(S) = \log\big(NAV + PnL(S)\big)$$

This accounts for more wealth certainly in the long run and for the investor's risk aversion, as the logarithmic function increases at a decreasing rate. This property encourages portfolio diversification and penalizes large drawdowns, making it suitable for long-term growth optimization. Our approach operates under the real-world probability measure

[2] Lleo, S. Advanced Portfolio Management Part 10 (p. 139-148), Certificate in Quantitative Finance. CQF Institute.

$\mathbb{P}$, acknowledging that the Kelly portfolio is ideally conceptualized within this measure. In later sections of the code, the Black-Litterman framework is modified to optimize the portfolio weights using this utility function rather than the standard mean-variance utility.

### 4.1 The Black-Litterman Model

The Black-Litterman model is a popular method for portfolio optimization that expresses a portfolio manager's views on the market and reflected in the portfolio allocation, removes the difficulties associated with estimating parameters such as expected returns on universal assets.

The model uses a Bayesian approach to combine the subjective views of an investor regarding the expected returns of one or more assets with the market equilibrium vector of expected returns (the prior distribution) to form a new, mixed estimate of expected returns. The resulting new vector of returns (the posterior distribution) leads to intuitive portfolios with sensible portfolio weights.

Implementing the Black-Litterman model involves choosing the equilibrium returns (allocations from the CAPM model or allocations in an index), updating the equilibrium returns with the views, allowing some uncertainty ($\tau \sum$), and finally apply the optimization expression:

$$\underbrace{argmax}_{w} \left\{ w' \hat{\mu} - \frac{1}{2} \lambda w' \sum w \right\}$$

$$\frac{\partial}{\partial w'} \left\{ w' \hat{\mu} - \frac{1}{2} \lambda w' \sum w \right\} = \hat{\mu} - 2 \lambda \sum w = 0$$

Solving for $w$:

$$w = \frac{1}{2 \lambda} \sum^{-1} \hat{\mu}$$

Black-Litterman insights:

- Expected Returns are changed to Equilibrium Returns $\hat{\mu} \rightarrow \pi$
- Expected Returns are updated with views
- Posterior is computed from prior $\pi \rightarrow \mu_{BL}$
- $\tau \sum$ gives the standard error of the estimate representing uncertainty
- Updated Equilibrium Returns (the posterior) overshadow Covariance input

To implement the Black-Litterman model, we first calculate market-implied equilibrium returns by deriving market capitalization weights using specialized indices such as MSCI, S&P Dow Jones, and ACWI. These indices serve as proxies for the market portfolio, and their constituent asset weights reflect the market's expectations. Then we calculate the expected returns and covariance of the ETFs within our portfolio under $\mathbb{P}$ using historical daily data from 2020 to 2024. The market equilibrium expected returns represent the 'prior' in the Black-Litterman framework. We then adjust these equilibrium returns based on investor views, blending market perspectives with subjective opinions to generate a new set of expected returns that reflect both market data and investor insights.

In portfolio theory and asset pricing, accurately estimating the expected return of the portfolio - 'getting the mean right' - is crucial[1] for determining optimal allocations. The preparation of datasets for the Black-Litterman model involves fetching daily data from Yahoo Finance using *yahooquery* and Kavout.com to compute market capitalizations and extract key financial metrics. This data is used to calculate the necessary inputs for portfolio optimization, including expected returns, covariance matrices, and market capitalization weights.

Lleo, S opined that in terms of dataset frequency daily data tend to be too noisy[3] for long term portfolio decisions. The risk-free rate, used in the Capital Asset Pricing Model (CAPM), is the return on the 13-week US Treasury Bill, retrieved from Yahoo Finance's API. This rate is used to calculate excess returns, ensuring our analysis is risk-adjusted appropriately. In the Fama French factor analysis section, we switch to use monthly data and derived returns to smooth out short-term volatility and enhance the reliability of our estimates.

The complexity of building the required inputs is due in part to the model's advanced nature and its reliance on both subjective and objective data. The challenge is in the detailed and accurate construction of these inputs:

1. Market Equilibrium Returns (Prior Distribution): In the Black-Litterman model, Market Equilibrium Returns, serving as the Prior Distribution, are derived using a reverse optimization method. The approach starts with the current market capitalization weights of assets in a well-diversified market portfolio, such as a global index.

    - These market weights are assumed to represent the market equilibrium, reflecting the collective expectations and valuations of all market participants. The market capitalization weights of the assets are $w_a$ in the formula.

    - By applying reverse optimization, we extract a set of Implied Excess Equilibrium Return Vector (Π) by working backwards from these market weights, using a covariance matrix of the asset returns (∑), scaled by the risk aversion coefficient ($\lambda$), to determine the expected returns that would make these weights optimal under the mean-variance framework. Market Equilibrium Returns, denoted as Π, represent the expected returns of assets in a market that is in equilibrium. These returns are derived from the CAPM, adjusted for risk aversion. The formula for the implied equilibrium returns (Π) is:

$$\Pi = \lambda \Sigma w_a$$

    - These derived returns, therefore, act as the foundational 'prior' estimates of expected returns for the Black-Litterman model before any individual investor views or forecasts are integrated.

    - The risk aversion coefficient, denoted as λ, is a crucial parameter in the Black-Litterman model. It reflects the degree of risk aversion of the typical investor in the market. A higher value of λ indicates greater risk aversion, meaning that the investor requires a higher return for taking on additional risk. Conversely, a lower value of λ indicates lower risk aversion. In accordance with the work of Idzorek (2007), the risk aversion coefficient of the US market is roughly 3.07. Consequently, we utilize this value as the value of λ when obtaining Π in this project. This provides a standardized measure based on empirical research, aligning the model with market conditions. The 3.07 value can be considered as borderline high level of risk aversion. While values 2.5 to 3.0 are considered a moderate level of risk aversion. Generally, a number from 1 (lowest risk aversion) to 5 (highest risk aversion) is attributed to an investor[4].

2. Covariance Matrix of the Returns (∑): this matrix represents the expected variances and covariances (i.e., the relationships) among the returns of the assets in the portfolio (as variable C in the code). It is crucial for understanding the risk characteristics of the portfolio and for calculating the optimal asset allocation.

3. Investor's Views (Subjective Views): these are the investor's personal forecasts or opinions about the performance of certain assets or asset classes. Views can be absolute (e.g., "Stock A will return 10% next year") or relative (e.g., "Stock A will perform better than Stock B"). Each view must be quantified in terms of expected return and accompanied by a confidence level.

4. Confidence in Views (Tau): Tau ($\tau$) is a scalar that represents the investor's confidence in their views. It adjusts the impact of these views on the overall portfolio, with higher confidence leading to a greater deviation from the

[3] Lleo, S. Advanced Portfolio Management Part 10 (p. 174), Certificate in Quantitative Finance. CQF Institute.

[4] https://www.disnat.com/en/learning/trading-basics/dbulletin/the-risk-aversion-coefficient

market equilibrium.

5. Omega ($\Omega$) - Uncertainty in the Views: Omega is a diagonal matrix that quantifies the uncertainty or error associated with the investor's views. It plays a critical role in determining how much the investor's views will affect the final portfolio allocation.

**CAPM (Capital Asset Pricing Model) Excess Returns:** we calculate the CAPM (Capital Asset Pricing Model) excess returns relative to ACWI (as a global market proxy). These excess returns serve as a starting point for understanding how much return an asset should generate, given its risk profile, compared to the global market.

**Historical Expected Returns:** we calculate the historical expected returns to be compared to CAPM-based returns and the implied equilibrium returns from the Black-Litterman process.

**CAPM Portfolio Expected Returns:** we calculate the CAPM portfolio expected returns which is the expected returns for each ETF using its beta relative to the ACWI.

**Problems / Anomaly Encountered:** while processing historical expected returns, we noticed that there is an unusual spike in VIXY when data collection period is set using start_date='2010-01-01' (see the screenshot of VIXY daily returns below). We have since adjusted the period from 2020 to 2024 but inserted this validation check in the project code to confirm that there are no extreme values where daily returns are more than 2.5%. This is to prevent data errors from other financial data packages.

```
extreme_returns = etfs_df[etfs_df['Daily Returns'] > 2.5]
print(extreme_returns)
```

```
                 Open       High        Low      Close  Adj Close  Volume  \
Date
2011-01-04 158280.000 162700.000 158280.000 158460.000 158460.000      56

             Market Cap   ETF  Daily Returns  Excess Returns
Date
2011-01-04 95020000.000  VIXY       1734.027        1734.027
```

The equilibrium returns Π calculated in the code serve as a baseline, which can be adjusted by incorporating specific views and confidence levels using the Black-Litterman model. This results in adjusted returns that better reflect the investor's views and market conditions, leading to more optimal portfolio allocations when combined with the quadratic optimization routine.

### 4.2 Systematic Backtesting of the 20 ETFs Asset Returns vs. the Market (SPY)

In this section, we calculate and visualize the rolling beta and alpha for the 20 ETFs relative to the market benchmark (SPY) and conduct systematic backtesting before proceeding with portfolio optimization. These metrics help us evaluate how the sensitivity (beta) and excess returns (alpha) of each ETF fluctuate over time in relation to the market. In the code implementation, we use a systematic approach to select an appropriate choice of the rolling window size (e.g., 30, 60 or 90 days) based on the Mean Squared Error (MSE) for the given window sizes.

The MSE evaluates how well the rolling beta for that window size explains the relationship between the ETF and SPY, such that a lower MSE values indicate that the window size is more appropriate for accurately estimating the relationship between the ETF and SPY.

As a result of looping through the different window sizes, the lowest MSE is selected as the optimal window for calculating the rolling beta and alpha:

```
Window Size: 30, MSE: 1.4879529149679669
Window Size: 60, MSE: 1.4742153707194299
Window Size: 90, MSE: 1.4774723838335575
Optimal Window Size: 60 with MSE: 1.4742153707194299
```

Rolling window size for backtesting asset returns vs the market shall be set at 60 days. Here are the generated Rolling Beta and Rolling Alpha plots that provide insights into the changing relationship between each ETF and the benchmark (e.g., SPY) over a rolling window of 60 days. The beta and alpha can be recalculated periodically as new data points are added, showing how the sensitivity of the ETF to the benchmark evolves over time.

The Rolling Beta plots presented for each ETF depict the time-varying sensitivity of these ETFs to the benchmark (likely SPY) over a rolling window period. The beta measures the volatility of each ETF relative to the market, with a beta of 1 indicating that the ETF moves in line with the market; higher than 1 is that ETF is more volatile than the market; and lower than 1 is low volatility (see Figure 2 and Table 4):

- Growth and Technology ETFs (VUG, MTUM, XLK) have betas higher than 1, are more volatile and sensitive to market trends, offering higher risk and higher reward potential.
- Value and Dividend ETFs (VTV, SPHD) tend to have lower betas, so more of a conservative nature and reduced sensitivity to market fluctuations.
- Safe-haven and Bond ETFs (SCHO, TLT, GLD) show low or negative beta, indicating their role as defensive assets that provide stability and protection during market volatility.
- Sector-specific ETFs (Energy, Oil, and Technology) exhibit volatile beta trends, reflecting their sensitivity to global events and specific market conditions.

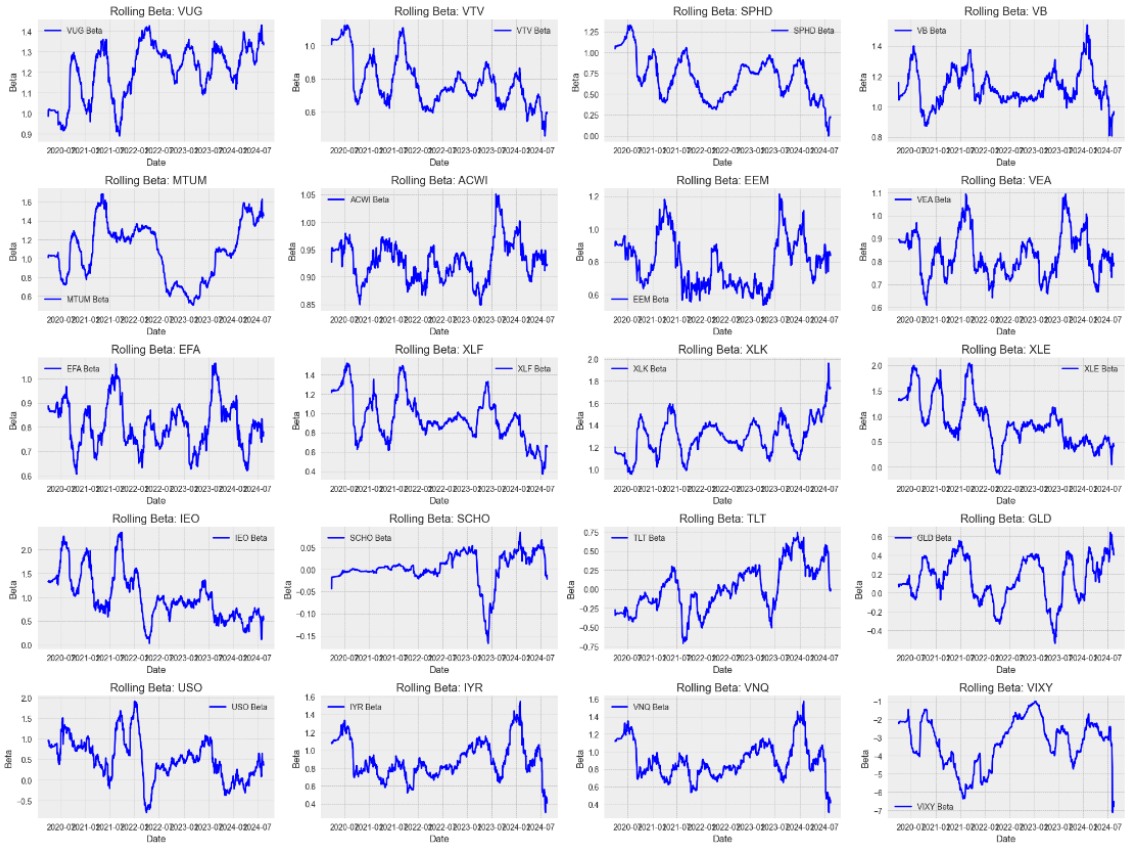



Figure 2: Rolling Beta for the 20 ETFs

Summary of the rolling beta plots:

- Growth and Technology ETFs (VUG, MTUM, XLK) have betas higher than 1, are more volatile and sensitive to market trends, offering higher risk and higher reward potential.
- Value and Dividend ETFs (VTV, SPHD) tend to have lower betas, so more of a conservative nature and reduced sensitivity to market fluctuations.
- Safe-haven and Bond ETFs (SCHO, TLT, GLD) show low or negative beta, indicating their role as defensive assets that provide stability and protection during market volatility.
- Sector-specific ETFs (Energy, Oil, and Technology) exhibit volatile beta trends, reflecting their sensitivity to global events and specific market conditions.

Table 4: Rolling Beta Analysis of each ETF

| ETF | Range | Evaluation Comments |
|---|---|---|
| **VUG** | 1.0 to 1.4 | Growth-focused ETF with higher volatility than the market. Peaks in 2021 show increased sensitivity to market changes. |
| **VTV** | 0.8 to 1.0 | Value ETF with lower volatility, indicating it is a more conservative investment with less market sensitivity. |
| **SPHD** | 0.5 to 1.25 | Fluctuating beta; high dividend ETF with periods of increased volatility in 2021 and 2023. |
| **VB** | 1.0 to 1.6 | Small-cap ETF with consistently high volatility, reflecting the increased risk associated with small-cap stocks. |
| **MTUM** | 0.6 to 1.4 | Momentum ETF with significant fluctuations in beta, reflecting its sensitivity to trend-following strategies. |
| **ACWI** | 0.9 to 1.1 | Global ETF with stable beta around 1, indicating alignment with global market movements. |
| **EEM** | 0.9 to 1.4 | Emerging markets ETF with high volatility, reflecting the instability and risk in emerging market investments. |
| **VEA** | 0.9 to 1.1 | Developed markets ETF with beta near 1, showing stability and alignment with broader market movements. |
| **EFA** | 0.85 to 1.2 | International ETF showing moderate volatility and stable performance relative to the market. |
| **XLF** | 1.0 to 1.4 | Financial sector ETF with beta above 1, reflecting heightened volatility due to sensitivity to interest rates and economic factors. |
| **XLK** | 1.0 to 2.0 | Technology sector ETF with a high and rising beta, indicating increased sensitivity and volatility, especially in 2022 and 2023. |
| **XLE** | 0.5 to 2.0 | Energy sector ETF with extreme fluctuations, driven by oil price movements and global energy market volatility. |
| **IEO** | 1.0 to 2.0 | Oil & gas ETF with consistently high beta, reflecting the volatility in the oil and gas sector. |
| **SCHO** | -0.1 to 0.1 | Short-term Treasury bond ETF with very low beta, showing minimal correlation with the market and stability as a low-risk asset. |
| **TLT** | -0.75 to 0.75 | Long-term Treasury bond ETF with fluctuating beta, often negative, reflecting an inverse relationship with market movements. |
| **GLD** | -0.25 to 0.25 | Gold ETF with beta close to 0, reflecting gold's role as a safe-haven asset with minimal correlation to the stock market. |
| **USO** | 0.5 to 1.5 | Crude oil ETF with high volatility, reflecting the sensitivity of oil prices to geopolitical events and market dynamics. |
| **IYR** | 0.6 to 1.1 | Real estate ETF with moderate volatility, showing that real estate assets are relatively stable but still sensitive to market movements. |
| **VNQ** | 0.8 to 1.6 | Real estate ETF with higher volatility than IYR, suggesting increased sensitivity to market changes in the real estate sector. |
| **VIXY** | -2.5 to -0.5 | Volatility ETF with a strong negative beta, indicating an inverse relationship with the market (rises in volatility during market declines). |

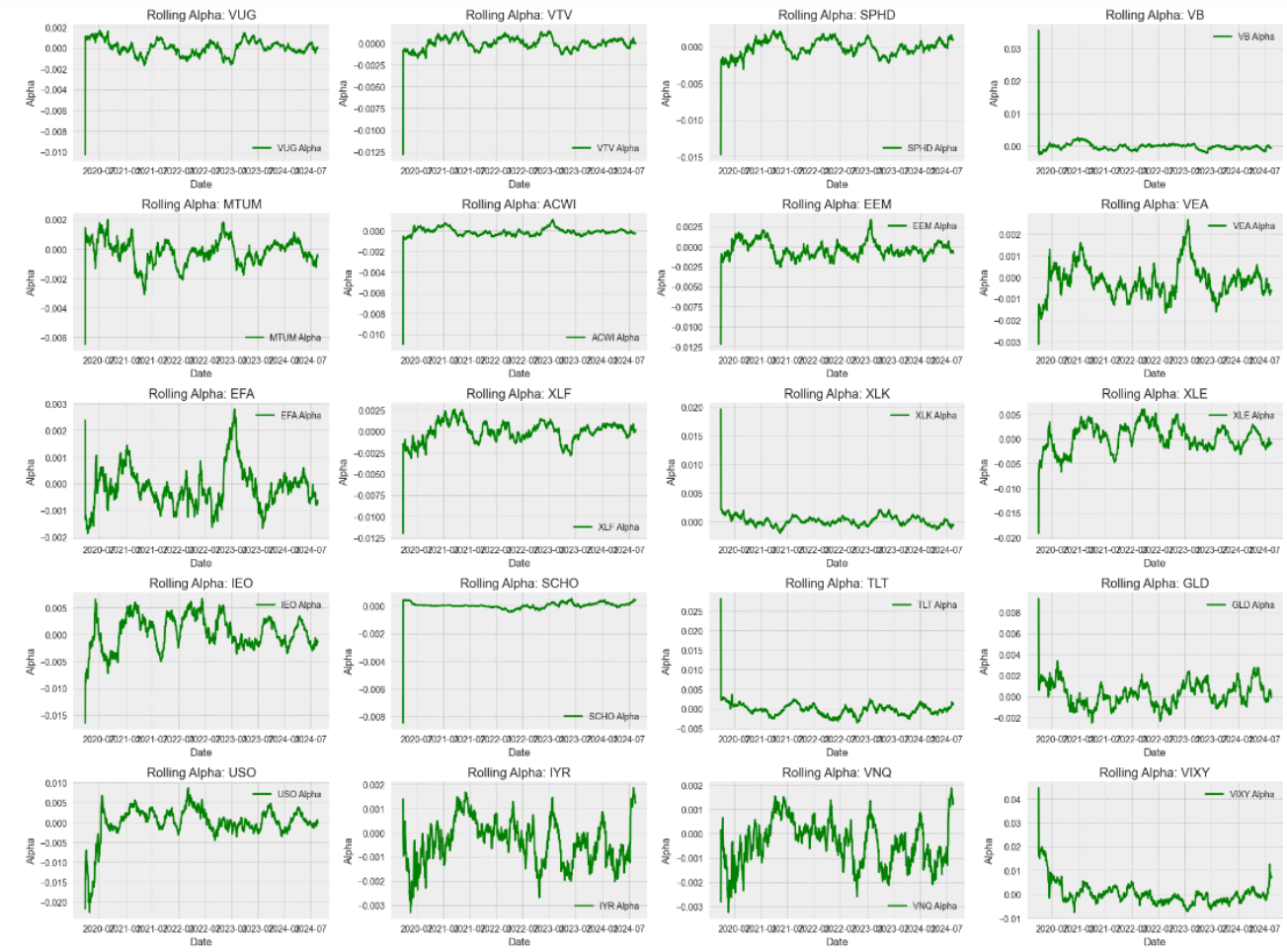



Figure 3: Rolling Alpha for the 20 ETFs

Alpha represents the excess return of the asset relative to the benchmark. Positive alpha means the ETF has outperformed the benchmark after accounting for its beta, while negative alpha suggests underperformance. The rolling alpha plots show how the ETF's alpha changes over time. This can reveal whether the ETF consistently outperforms or underperforms the benchmark, and how external factors or management decisions affect performance, e.g. stable alpha, positive alpha trend, or negative alpha trend (see Table 5) .

Summary of the rolling alpha plots:

- Outperforming ETFs: SPHD, VB, GLD, and VIXY exhibit positive alpha, suggesting consistent outperformance relative to the benchmark.
- Underperforming ETFs: VTV, XLF, and XLK show negative alpha, indicating steady underperformance over the period.
- Volatile Alpha: XLE, USO, and IEO have highly variable alpha, showing periods of both significant outperformance and underperformance, likely driven by sector-specific factors like oil prices.

Table 5: Rolling Alpha Analysis of each ETF

| ETF | Range | Evaluation Comments |
|---|---|---|
| **VUG** | -0.004 to 0.002 | Growth ETF with slightly negative alpha over time, indicating that it generally underperformed the market slightly, but with occasional small outperformance. |
| **VTV** | -0.010 to -0.002 | Value ETF with consistent negative alpha, suggesting consistent underperformance relative to the benchmark. Reflects lower market outperformance. |
| **SPHD** | 0.000 to 0.005 | High Dividend ETF with positive alpha, indicating consistent outperformance of the benchmark. Dividends might contribute to the outperformance. |
| **VB** | 0.000 to 0.030 | Small-Cap ETF with significant positive alpha, suggesting strong outperformance of the market, particularly in 2022 and early 2023. |
| **MTUM** | -0.004 to 0.003 | Momentum ETF with fluctuating alpha, showing periods of outperformance and underperformance. The strategy is sensitive to market trends. |
| **ACWI** | -0.010 to 0.000 | Global ETF with slightly negative alpha, indicating slight underperformance relative to the global market benchmark. |
| **EEM** | -0.002 to 0.005 | Emerging Markets ETF with periods of positive alpha, reflecting outperformance during certain periods, though often close to the benchmark return. |
| **VEA** | -0.010 to 0.010 | Developed Markets ETF showing alpha swings. The ETF shows outperformance in some periods but also underperforms significantly at times. |
| **EFA** | -0.002 to 0.003 | International ETF with fluctuations in alpha, showing some periods of outperformance but generally close to market returns. |
| **XLF** | -0.0125 to 0.0025 | Financial Sector ETF with consistent negative alpha, indicating a steady underperformance relative to the market, likely influenced by economic factors. |
| **XLK** | -0.004 to 0.002 | Technology Sector ETF with negative alpha, indicating that despite market volatility, the ETF underperformed the benchmark during the majority of the period. |
| **XLE** | -0.010 to 0.015 | Energy Sector ETF with significant alpha variability, indicating periods of outperformance (in energy boom periods) and underperformance in others. |
| **IEO** | -0.015 to 0.005 | Oil & Gas ETF with fluctuating alpha, indicating volatility and periods of outperformance and underperformance relative to the market. |
| **SCHO** | 0.000 to 0.000 | Short-term Treasury Bond ETF with flat alpha, indicating that it closely tracks its benchmark, reflecting its low-risk and low-return nature. |
| **TLT** | -0.005 to 0.015 | Long-term Treasury Bond ETF with slightly positive alpha, indicating periods of outperformance during times of market stress or rate changes. |
| **GLD** | 0.000 to 0.015 | Gold ETF with consistently positive alpha, reflecting gold's status as a safe haven asset and periods of outperformance during market uncertainty. |
| **USO** | -0.025 to 0.010 | Crude Oil ETF with highly fluctuating alpha, reflecting the volatility in the oil market and periods of significant outperformance or underperformance. |
| **IYR** | -0.002 to 0.002 | Real Estate ETF with slightly positive alpha, indicating that real estate investments generally track the market but have had brief periods of outperformance. |
| **VNQ** | -0.002 to 0.003 | Real Estate ETF with slightly positive alpha, suggesting a relatively stable performance with mild outperformance during certain periods. |
| **VIXY** | 0.000 to 0.030 | Volatility ETF with highly positive alpha, indicating strong outperformance during market downturns, as it captures the rising volatility during market stress. |

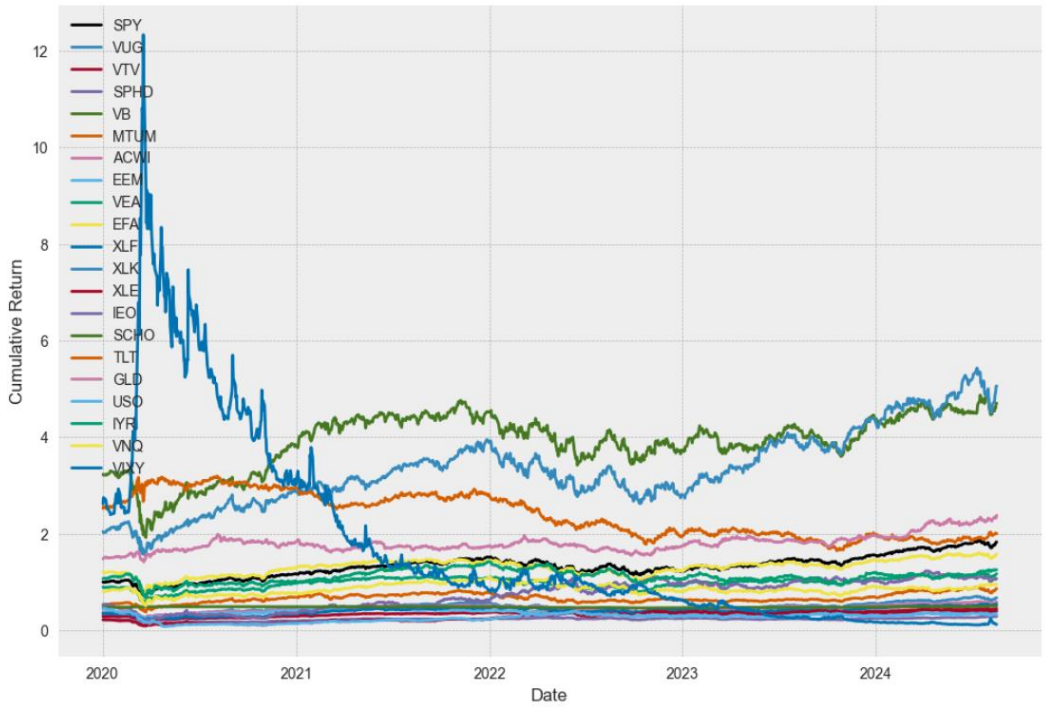
Next, we compute key performance metrics and P&L returns for each ETF compared to the market (SPY). This

includes calculating Sharpe Ratio, Sortino Ratio, Maximum Drawdown, Annualized Return, Volatility, and Cumulative P&L returns for each ETF.

Key metrics:

1. Sharpe Ratio: Risk-adjusted return, higher is better.
2. Sortino Ratio: Like Sharpe, but penalizes only downside risk, focusing on harmful volatility.
3. Maximum Drawdown: The worst peak-to-trough decline, of which a lower value is better.
4. Annual Return: The ETF's average return, annualized.
5. Volatility: The ETF's risk (standard deviation of returns), annualized.
6. Cumulative Returns (P&L): The total profit or loss of the ETF over time.

Figure 4: Cumulative P&L returns for 20 ETFs vs SPY




Performance Metrics Comparison (ETFs vs SPY):

| | ETF | Annual Return | Volatility | Sharpe Ratio | Sortino Ratio | Max Drawdown |
|---|---|---|---|---|---|---|
| 0 | VB | 0.598 | 1.068 | 0.550 | 2.161 | -0.420 |
| 1 | XLK | 0.463 | 0.563 | 0.804 | 1.554 | -0.336 |
| 2 | TLT | 0.297 | 0.732 | 0.392 | 1.590 | -0.484 |
| 3 | GLD | 0.219 | 0.271 | 0.770 | 1.304 | -0.220 |
| 4 | IEO | 0.144 | 0.481 | 0.278 | 0.249 | -0.664 |
| 5 | EFA | 0.124 | 0.226 | 0.505 | 0.525 | -0.339 |
| 6 | IYR | 0.083 | 0.260 | 0.281 | 0.265 | -0.423 |
| 7 | VEA | 0.057 | 0.211 | 0.221 | 0.200 | -0.353 |
| 8 | VUG | 0.046 | 0.407 | 0.087 | 0.065 | -0.356 |
| 9 | MTUM | 0.034 | 0.330 | 0.073 | 0.058 | -0.341 |
| 10 | XLE | 0.034 | 0.521 | 0.045 | 0.036 | -0.606 |
| 11 | VNQ | 0.029 | 0.277 | 0.070 | 0.063 | -0.424 |
| 12 | XLF | -0.010 | 0.404 | -0.049 | -0.038 | -0.429 |
| 13 | ACWI | -0.018 | 0.354 | -0.079 | -0.058 | -0.335 |
| 14 | VIXY | -0.020 | 1.089 | -0.027 | -0.051 | -0.991 |
| 15 | VTV | -0.036 | 0.391 | -0.118 | -0.086 | -0.368 |
| 16 | SPHD | -0.082 | 0.430 | -0.213 | -0.156 | -0.414 |
| 17 | USO | -0.082 | 0.516 | -0.178 | -0.145 | -0.838 |
| 18 | SCHO | -0.102 | 0.245 | -0.455 | -0.300 | -0.057 |
| 19 | EEM | -0.103 | 0.373 | -0.303 | -0.231 | -0.398 |
| 20 | SPY | 0.154 | 0.215 | 0.669 | 0.630 | -0.337 |

**Evaluation of Performance Metrics 20 ETFs vs SPY** (see Table 6 in next page for detailed key observations on the comparisons):

**Top Performers:** VB, XLK, TLT, and GLD had significantly higher returns compared to SPY with decent or better risk-adjusted metrics.

**Underperformers:** ETFs like XLF, ACWI, VIXY, USO, VTV, SPHD, and EEM showed negative returns, under-performing SPY across multiple metrics, often with higher volatility and deeper drawdowns.

**Defensive Assets:** GLD and TLT performed better than SPY in terms of downside protection (Sortino ratio), and their returns were 1.4x to 1.9x higher than SPY's.

Table 6: Key Observations of 20 ETFs vs the Market SPY Benchmark on Annual Return, Volatility, Sharpe Ratio, Sortino Ratio, and Max Drawdown (derived from Performance Metrics). SPY metrics are at the last row.

| ETF | ANNUAL RETURN COMPARISON | VOLATILITY COMPARISON | SHARPE RATIO COMPARISON | SORTINO RATIO COMPARISON | MAX DRAWDOWN COMPARISON | KEY OBSERVATIONS |
|---|---|---|---|---|---|---|
| **VB** | 3.9x higher than SPY | 5x higher than SPY | 17.5% lower than SPY | 3.5x higher than SPY | 25% deeper than SPY | High returns and risk, strong Sharpe and Sortino, a top performer. |
| **XLK** | 3x higher than SPY | 2.6x higher than SPY | 20% higher than SPY | 2.5x higher than SPY | 1x lower than SPY | Excellent risk-adjusted performance, high Sharpe ratio. |
| **TLT** | 1.9x higher than SPY | 3.4x higher than SPY | 41.4% lower than SPY | 2.5x higher than SPY | 44% deeper than SPY | Defensive asset with low returns, high volatility, but decent downside risk management. |
| **GLD** | 1.4x higher than SPY | 1.3x higher than SPY | 12.7% higher than SPY | 2x higher than SPY | 35% lower than SPY | Defensive, low volatility, good risk-adjusted returns, portfolio stabilizer. |
| **IEO** | Similar to SPY | 2.2x higher than SPY | 58.2% lower than SPY | 4x lower than SPY | 97% deeper than SPY | High drawdown and poor Sharpe/Sortino ratios, risky energy sector exposure. |
| **EFA** | 18.8% lower than SPY | Similar to SPY | 24.4% lower than SPY | 16% lower than SPY | Similar to SPY | Low returns but stable, moderate risk-adjusted performance. |
| **IYR** | 1.9x lower than SPY | 1.2x higher than SPY | 2.5x lower than SPY | 2.5x lower than SPY | 26% deeper than SPY | Low returns with moderate volatility, but significant drawdown. |
| **VEA** | 2.7x lower than SPY | Similar to SPY | 2.9x lower than SPY | 3x lower than SPY | Similar to SPY | Low returns and risk-adjusted metrics, underperforms SPY. |
| **VUG** | 3.4x lower than SPY | 1.9x higher than SPY | 7.7x lower than SPY | 9.7x lower than SPY | Similar to SPY | Underperforms with low Sharpe and Sortino ratios. |
| **MTUM** | 4.5x lower than SPY | 1.5x higher than SPY | 9.1x lower than SPY | 10.6x lower than SPY | Similar to SPY | Underperforms SPY with poor risk-adjusted metrics. |
| **XLE** | 4.5x lower than SPY | 2.4x higher than SPY | 14.5x lower than SPY | 17x lower than SPY | 80% deeper than SPY | Energy sector struggles, high volatility, and drawdown. |
| **VNQ** | 5x lower than SPY | 1.3x higher than SPY | 9.4x lower than SPY | 9.8x lower than SPY | 26% deeper than SPY | Low returns with significant drawdowns, weak risk-adjusted metrics. |
| **XLF** | Negative | 1.9x higher than SPY | Negative | Negative | 27% deeper than SPY | Underperforms with negative returns and poor risk-adjusted performance. |
| **ACWI** | Negative | 1.6x higher than SPY | Negative | Negative | Similar to SPY | Global ETF underperforms SPY with negative returns. |
| **VIXY** | Negative | 5x higher than SPY | Negative | Negative | 2.9x deeper than SPY | Extremely volatile, large drawdown, performs poorly in calm markets. |
| **VTV** | Negative | 1.8x higher than SPY | Negative | Negative | 9.2% deeper than SPY | Value ETF underperforms SPY with negative risk-adjusted returns. |
| **USO** | Negative | 2.4x higher than SPY | Negative | Negative | 2.5x deeper than SPY | Poor performance and very high drawdowns, extreme volatility in oil markets. |
| **SPHD** | Negative | 2x higher than SPY | Negative | Negative | 23% deeper than SPY | Poor performance, with significant downside risk and drawdown. |
| **SCHO** | Negative | Similar to SPY | Negative | Negative | 83% lower than SPY | Short-term bonds perform poorly with minimal risk-adjusted returns. |
| **EEM** | Negative | 1.7x higher than SPY | Negative | Negative | 18% deeper than SPY | Emerging markets underperform, with high drawdown and poor risk-adjusted performance. |
| **SPY** | **0.154** | **0.215** | **0.667** | **0.628** | **-0.337** | SPY is the benchmark for comparison. |

### 4.3 Initial Market Weights with Market Capitalization ($W$) and Adjusted Weights with 10% Capping ($Wa$)

After aggregating the latest market capitalization for each ETF asset, we calculate a simple market weights based on capitalization as follows:

$$w_{mkt} = \frac{Market\ Cap_{\ ETF}}{\sum_{i=1}^{n} Market\ Cap_{\ i}}$$

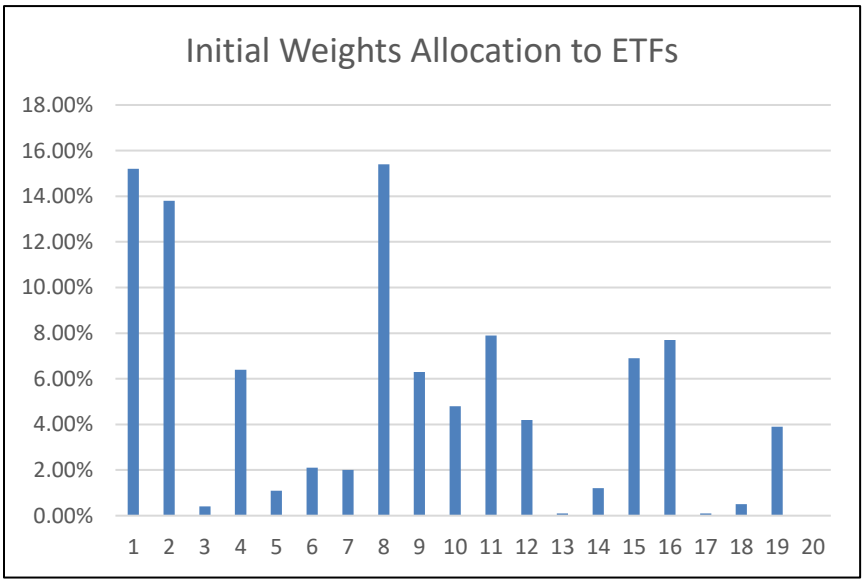
Where:

- $w_{mkt}$ is the market weight of each asset ETF.
- $Market\ Cap_{\ ETF}$ is the market capitalization of the particular ETF.
- n is the total number of ETFs in the portfolio (at this stage there are 20 ETFs).

Referring to the computed initial market weights based on market capitalization, we observe high weight allocations to VEA, VTV, and VUG, each with a weight exceeding 10% (see Table 7). This high concentration skews the portfolio towards these ETFs, introducing concentrated risk, which conflicts with standard diversification principles.

Table 7: Initial Market Weights (W) Allocation to 20 ETFs

| ETF | Initial Weight | ETF | Initial Weight |
|---|---|---|---|
| VUG | 15.2% | XLK | 7.9% |
| VTV | 13.8% | XLE | 4.2% |
| SPHD | 0.4% | IEO | 0.1% |
| VB | 6.4% | SCHO | 1.2% |
| MTUM | 1.1% | TLT | 6.9% |
| ACWI | 2.1% | GLD | 7.7% |
| EEM | 2.0% | USO | 0.1% |
| VEA | 15.4% | IYR | 0.5% |
| EFA | 6.3% | VNQ | 3.9% |
| XLF | 4.8% | VIXY | 0.0% |




Referring to industry best practices (S&P Dow Jones Indices) on weight allocations[5], we apply a capping rule to ensure better diversification and reduce concentration risk (see Figure 5). The following steps are taken:

1. Cap Individual ETF Weights: We ensure that no single ETF exceeds 10% of the portfolio to prevent overexposure to any single asset.

2. Redistribute Excess Weight: The excess weight above 10% is redistributed proportionally to the remaining ETFs, ensuring that the overall portfolio remains well-diversified while keeping the total weight sum at ~100%.

[5] S&P U.S. Style Indices Methodology, Daily Capped Growth and Daily Capped Value Indices. July 2024. p10: "If any company's weight exceeds 10%, that company's weight is capped at 10% and all excess weight is proportionally redistributed to all uncapped companies within the index."

```
Initial Market Weights:

ETF
VUG    0.152
VTV    0.138
SPHD   0.004
VB     0.064
MTUM   0.011
ACWI   0.021
EEM    0.020
VEA    0.154
EFA    0.063
XLF    0.048
XLK    0.079
XLE    0.042
IEO    0.001
SCHO   0.012
TLT    0.069
GLD    0.077
USO    0.001
IYR    0.005
VNQ    0.039
VIXY   0.000
Name: Market Cap, dtype: float64
```

```
VUG original weight: 0.1517
New weight for VUG: 0.1000, Excess weight: 0.0517
VTV original weight: 0.1459
New weight for VTV: 0.1000, Excess weight: 0.0459
SPHD original weight: 0.0043
VB original weight: 0.0720
MTUM original weight: 0.0124
ACWI original weight: 0.0230
EEM original weight: 0.0222
VEA original weight: 0.1724
New weight for VEA: 0.1000, Excess weight: 0.0724
EFA original weight: 0.0765
XLF original weight: 0.0585
XLK original weight: 0.0963
XLE original weight: 0.0517
IEO original weight: 0.0009
SCHO original weight: 0.0148
TLT original weight: 0.0835
GLD original weight: 0.0934
USO original weight: 0.0016
IYR original weight: 0.0058
VNQ original weight: 0.0480
VIXY original weight: 0.0001
Sum of Final Weights: 0.9999999999999999
```

Figure 5: Redistribution of Excess Weight from Initial Market Weights

Once the capped weights are established, we perform a risk adjustment using the covariance matrix (Σ) to further refine the portfolio. This step involves optimizing the portfolio based on both the adjusted returns (using a model like Black-Litterman) and the covariance matrix to minimize risk while adhering to the new weight constraints. While market cap weighting is commonly used in long-term, buy-and-hold strategies, it may not be ideal for active management.

### 4.4 Correlation Analysis and Portfolio Choice

Our portfolio selection approach follows *Option A* as outlined in the given Project QA. Out of the twenty-two (22) assets and risk-free rate, we compute initial weights based on market capitalization of the 18 Factors, 1 Benchmark (ACWI), and 1 Emerging Markets (EEM) to make up total twenty (20) assets, which excludes the SPY benchmark and RFR. Using the correlation heatmap analysis of these assets, we narrow down the Portfolio Choice to eight (8) 'optimal assets' using a heuristically determined correlation threshold. The heatmap is shown here in Figure 6:

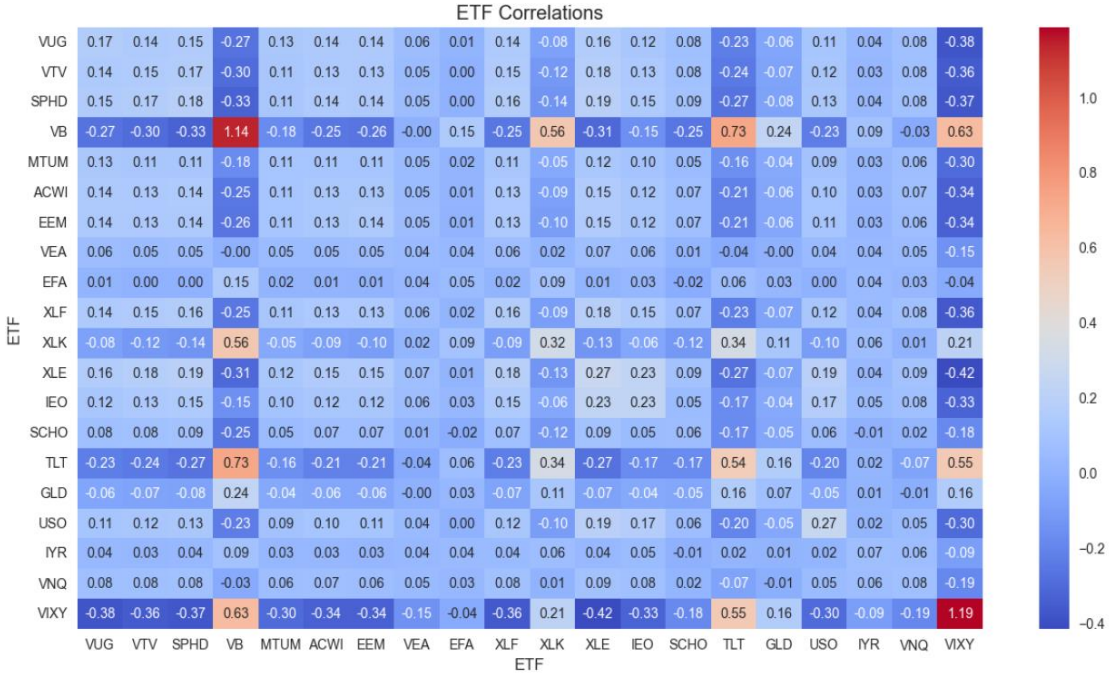



Figure 6: ETF correlations heatmap visually represents the correlation coefficients between different ETFs.

**ETFs with positive correlations:**

1. VB and TLT (0.73): This strong positive correlation indicate that the Small Cap ETF (VB) and Treasury Bonds (TLT) move together to a significant extent. This is an unusual combination since equities and bonds typically have a lower correlation, which may imply specific market conditions driving these two assets together.

2. VIXY and TLT (0.55): This suggests a positive relationship between volatility and Treasuries, which is expected, as Treasuries are often seen as a safe haven during periods of heightened volatility.

3. VB and XLK (0.56): A moderate positive correlation between Small Cap and Technology ETFs indicates that these two sectors share some common market drivers, likely linked to growth trends.

4. VIXY and VB (0.63): The positive correlation between volatility and small caps suggests that during periods of market stress, small-cap stocks (VB) are sensitive to volatility changes, aligning with the expectation that smaller companies are more exposed to market fluctuations.

**ETFs with negative correlations:**

1. VIXY and XLE (-0.42): The volatility index (VIXY) is negatively correlated with Energy (XLE), which may suggest that energy stocks behave differently in times of market stress or heightened volatility, making energy a potential diversification tool.

2. VIXY and VUG (-0.38): The volatility index (VIXY) is negatively correlated with growth stocks (VUG), implying that growth stocks tend to underperform when market volatility increases, consistent with their typical behavior as higher-risk assets.

3. VIXY and SPHD (-0.37): A negative correlation between volatility and the High Dividend Low Volatility ETF (SPHD) indicates that dividend-paying stocks, known for their stability, tend to perform better in low-volatility environments, highlighting SPHD's potential as a defensive asset.

**ETFs with near-zero correlations:**

1. VEA with multiple ETFs: The Developed Market ETF (VEA) shows low correlations with many other ETFs, implying that it can serve as a good diversifier when combined with more highly correlated assets in the portfolio. Low correlations suggest that international exposure may reduce the overall risk of the portfolio.

**Sector-specific observations:**

1. XLE (Energy): XLE exhibits low correlations with ETFs from other sectors, such as technology (XLK) and financials (XLF). This means that energy stocks could be strategically included in the portfolio to reduce sector-specific risk.

2. GLD (Gold): GLD behaves differently from most equity ETFs, reinforcing its role as a potential safe haven or hedge. Gold is often used as a store of value during times of market turmoil, which is reflected in its distinct correlation pattern compared to equity ETFs.

Based on the correlation analysis, we can make decisions about which ETFs to retain for diversification and risk reduction so as to narrow down to less than ten (10) ‘optimal bets. We remove highly correlated ETFs, namely VB, VTV, TLT, and VIXY. While ensuring exposure to key sectors, regions, and asset types, we further remove EEM, VEA, EFA, XLF, SCHO, USO, IYR, and VNQ from the list.

This leaves us with eight (8) optimal bets of a more concentrated and diversified portfolio, with exposure to a mix of growth stocks (VUG), technology (XLK), energy (XLE), gold (GLD), and other less correlated sectors such as high dividend low volatility stocks (SPHD), momentum stocks (MTUM), global equities (ACWI), and oil & gas exploration (IEO), providing a balanced exposure across various asset classes and reducing overall portfolio risk.

After identifying the optimal ETFs based on the correlation analysis, we proceed to calculate the annualized expected returns (R) and the covariance matrix (C) by filtering the original asset returns to include only the optimal eight (8) ETFs present in the 'correlation_df' (see Table 8).

The covariance matrix (C) measures the covariance of returns between each pair of assets in the portfolio. It is fundamental in assessing the risk of the portfolio because it provides a measure of how assets move in relation to one another.

The covariance matrix and annualized expected returns derived from the filtered set of eight (8) ETFs represent an optimized portfolio choice:

1. The expected returns, ranging from 0.047 for VUG to 0.244 for GLD, highlight a mix of growth-oriented and defensive assets, balancing both risk and potential reward.

2. The covariance matrix captures the co-movement between these assets, indicating varying degrees of correlation, with some assets like XLK and XLE exhibiting negative covariance, which enhances diversification by reducing overall portfolio volatility.

Table 8: Covariance Matrix (C) and Annualized Expected Returns (R)

| ETF | Expected Returns |
|---|---|
| VUG | 0.047 |
| SPHD | -0.078 |
| MTUM | 0.034 |
| ACWI | -0.018 |
| XLK | 0.588 |
| XLE | 0.034 |
| IEO | 0.154 |
| GLD | 0.244 |

| Cov (C) | VUG | SPHD | MTUM | ACWI | XLK | XLE | IEO | GLD |
|---|---|---|---|---|---|---|---|---|
| **VUG** | 0.028 | 0.026 | 0.017 | 0.02 | -0.019 | 0.033 | 0.023 | -0.007 |
| **SPHD** | 0.026 | 0.034 | 0.016 | 0.022 | -0.034 | 0.043 | 0.03 | -0.009 |
| **MTUM** | 0.017 | 0.016 | 0.012 | 0.013 | -0.008 | 0.021 | 0.015 | -0.004 |
| **ACWI** | 0.02 | 0.022 | 0.013 | 0.016 | -0.017 | 0.028 | 0.02 | -0.006 |
| **XLK** | -0.019 | -0.034 | -0.008 | -0.017 | 0.101 | -0.038 | -0.015 | 0.017 |
| **XLE** | 0.033 | 0.043 | 0.021 | 0.028 | -0.038 | 0.073 | 0.059 | -0.01 |
| **IEO** | 0.023 | 0.03 | 0.015 | 0.02 | -0.015 | 0.059 | 0.054 | -0.006 |
| **GLD** | -0.007 | -0.009 | -0.004 | -0.006 | 0.017 | -0.01 | -0.006 | 0.005 |

The combination of these assets, with their diverse return profiles and correlation characteristics, forms a well-rounded portfolio that is positioned to capitalize on growth opportunities while managing risk through diversification.

### 4.5 Risk-Based Multi-Factor Construction

**Weight Allocation Strategies: t**here are three primary approaches to constructing multi-factor portfolios, each with its unique methodology and set of challenges:

1. Heuristic Multi-factor Construction offers a straightforward, intuitive method for assembling investment portfolios based on multiple factors such as value, momentum, quality, size, and volatility. Despite its simplicity, this approach suffers from significant drawbacks, including high parameter and factor selection biases due to

subjective judgment, potential overfitting, a lack of robustness in changing market environments, and higher tracking errors. These issues collectively lead to inefficiencies in generating alpha (excess returns over the benchmark) and result in suboptimal risk-adjusted performance, making it less favorable for achieving consistent, risk-adjusted investment returns.

2. Optimized Multi-factor Construction advances beyond the heuristic approach by incorporating a systematic risk framework and employing large-scale optimization techniques. It aims to align the portfolio's expected risk with a specific range, consistent with the benchmark's risk profile, and focuses on minimizing the expected tracking error. However, the construction of a robust optimization model is challenging, especially in balancing the reduction of tracking error with the maintenance of an acceptable level of risk. For example, tightening constraints can reduce tracking error but may also lead to a more concentrated portfolio, increasing the risk from specific securities.

3. Risk-Based Multi-factor Construction[6] prioritizes risk management in portfolio construction, utilizing advanced analytics to assess and mitigate various types of risks. This method focuses on optimizing the portfolio to balance returns with risk considerations, often through diversification across securities with different risk-return profiles. Although modeling and predicting risks accurately can be complex, this approach is valued for its ability to produce resilient portfolios capable of withstanding market volatility and uncertainties, offering investors a more stable performance over time.

In the section on factor data collection and analysis, we focus on selecting and weighting factors using the above list's 3rd strategy, which models portfolios based on factor risk premiums. This approach is particularly suited to portfolios that need to withstand market volatility and uncertainty. By prioritizing risk management, it allows for diversification across different risk-return profiles, reducing exposure to extreme market movements and providing more stable performance over time. Given the complexities involved in predicting risks accurately, this method is highly valued for its resilience in challenging market environments.

We apply this framework by calculating factor loadings using the Fama-French five factors' data[7] and model to capture exposures to key risk factors such as market (Mkt-RF), size (SMB), value (HML), profitability (RMW), and investment (CMA). These factors help to tilt the portfolio toward risk premiums that are associated with long-term returns while balancing against volatility. The Fama-French five factors are stored in 'F-F_Research_ Data_5_Factors_2x3_monthly.csv' based on the period from July 1963 to June 2024.

The Fama-French Five-Factor model (2015) is an extension of the Capital Asset Pricing Model (CAPM) developed by Eugene F. Fama and Kenneth R. French on the return (R) of the portfolio:

$$R = \alpha + \beta_m MKT + \beta_s SMB + \beta_h HML + \beta_r RMW + \beta_c CMA + \epsilon$$

Here, $\epsilon$ is the error term, representing the portion of returns not explained by the five factors; $\beta_m$, $\beta_s$, $\beta_h$, $\beta_r$, and $\beta_c$ are coefficients that measure the sensitivity of the return to each respective market factor; and $\alpha$ is the asset's expected return that is not explained by the risk factors in the model. The model aims to explain asset returns through five factors, providing a more comprehensive view than the CAPM, which only considers the market risk factor. This model is particularly useful for portfolio management and understanding the drivers behind asset returns.

The Kenneth R. French Data Library's datasets[8] are used to compute expected returns using a multi-factor model. In the downloaded CSV dataset, Rm-Rf includes all NYSE, AMEX, and NASDAQ firms. SMB, HML, RMW, and CMA for July of year t to June of t+1 include all NYSE, AMEX, and NASDAQ stocks for which they have market equity

[6] A Practitioner's Guide to Multi-Factor Portfolio Construction. Nasdaq Global Information Services. Lin, R. (2020).

[7] Kenneth French's online database: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

[8] Fama/French 5 Factors (2x3): https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_5_Factors_2x3_CSV.zip

data for December of t-1 and June of t, (positive) book equity data for t-1 (for SMB, HML, and RMW), non-missing revenues and at least one of the following: cost of goods sold, selling, general and administrative expenses, or interest expense for t-1 (for SMB and RMW), and total assets data for t-2 and t-1 (for SMB and CMA).

Table 9: Fama-French Five Factors

| Factor | Type | Expected Premium |
|---|---|---|
| MKT | Market Risk | Excess return of the market over the risk-free rate. Represents the compensation for taking on higher volatility of stocks over a risk-free asset. |
| SMB | Size<br>"Small Minus Big" | Excess return for investing in small-cap stocks over large-cap stocks. Historically small-cap stocks have tended to outperform large-cap stocks. |
| HML | Value<br>"High Minus Low" | Excess return for investing in value stocks over Growth stocks. Value stocks have high book-to-market ratios, while growth stocks have low book-to-market ratios. Value in essence buys assets with high yields (or low prices) and sells assets with low yields (or high prices) and is a pervasive factor across many different asset classes, can be implemented cheaply and in size by a large investor (Andrew Ang, 2014). Investors are compensated for the additional risk and potentially longer investment horizon associated with value investing. |
| RMW | Profitability<br>"Robust Minus Weak" | Excess return of companies with high operating profitability over those with low profitability. |
| CMA | Investment<br>"Conservative Minus Aggressive" | Excess return of companies with conservative low investment growth strategies over those with aggressive strategies (a risk factor associated with high investment expenditures). |

Once the dataset is loaded into the Python / Jupyter Notebook environment, we utilize pandas to manage and extract the data. The calculation of R is then performed taking into consideration both the sensitivity of the investment to each of these factors (captured by the beta coefficients) and the factor premiums. The model leverages historical data to understand the behavior of investment returns in relation to various market conditions to estimate the expected excess return over the risk-free rate for each period in the dataset.

Table 10: Other factors that capture various market anomalies and risk premia that are not fully explained by traditional asset pricing models:

| Factor | Type | Expected Premium |
|---|---|---|
| MOM | Momentum | Positive premium for stocks with strong past performance, on the premise that stocks that have performed well in the past continue to perform well in the near future. |
| ST Rev | Short-Term Reversal | Positive premium for recently underperforming stocks due to investor overreaction that have experienced extreme losses tend to reverse in the short term. |
| LT Rev | Long-Term Reversal | Positive premium for long-term underperforming stocks as a correction of mispricing that tend to reverse their performance. |
| Market Beta | Market Beta (β) | Higher returns for stocks with higher beta, according to CAPM, though empirically challenged by BAB. In essence, sensitivity of a stock's returns to the returns of the market portfolio. |
| BAB | Betting Against Beta | Positive premium for low-beta stocks, contrary to CAPM's prediction. This is strategy based on the finding that lower-beta stocks have higher risk-adjusted returns than higher-beta stocks. |

**Strategy of Dynamic Weight Adjustment Based on Factor Loadings**

We adjust the portfolio allocations by applying thresholds to the factor loadings, increasing or decreasing weights based on the significance of each factor exposure. This method ensures that the portfolio is optimized for factor risk premiums and adjusts based on the portfolio's sensitivity to these factors.

To collect and analyze data, we gather ETF data using Yahoo Finance! and load the Fama-French five-factor model from a local file, ensuring both datasets are aligned on a monthly basis. We then run regression analyses to measure the portfolio's returns against Fama-French factors, including Market (Mkt-RF), Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA), determining the portfolio's sensitivity (factor loadings) to these factors.

Positive factor loadings, such as on SMB, indicate a tilt towards small-cap stocks, while negative loadings imply an inverse relationship, such as a preference for large-cap over small-cap stocks. This analysis allows us to assess whether factors are contributing positively to the portfolio's returns or adding unwanted risk, aligning exposures with investor strategy. For example, if the analysis shows a significant positive loading towards small-cap stocks but a downturn in that segment is anticipated, reducing exposure to that factor may be beneficial. Similarly, increasing exposure to factors like HML (value) can help if value stocks are expected to perform well. In more advanced strategies or future work, we can use these insights to construct long-short portfolios that enhance desired factor exposures (e.g., positive exposure to value) while minimizing exposure to unwanted factors (e.g., small caps).

Regression Process:

1. Data Collection: We collect the Fama-French factors data and ETF price data, aligning them to perform regression analysis.
2. Regression: For each ETF, we run regression models using the Fama-French factors as explanatory variables, subtracting the risk-free rate from ETF returns to compute excess returns.
3. Interpretation: The factor loadings give insights into how sensitive each ETF is to each of the Fama-French factors, which we plot using scatter plots and regression lines for visualization.
4. Adjustment: Based on these factor loadings, we can adjust ETF allocations to align the portfolio better with investor goals, reducing exposure to unwanted factors and increasing exposure to favorable factors.

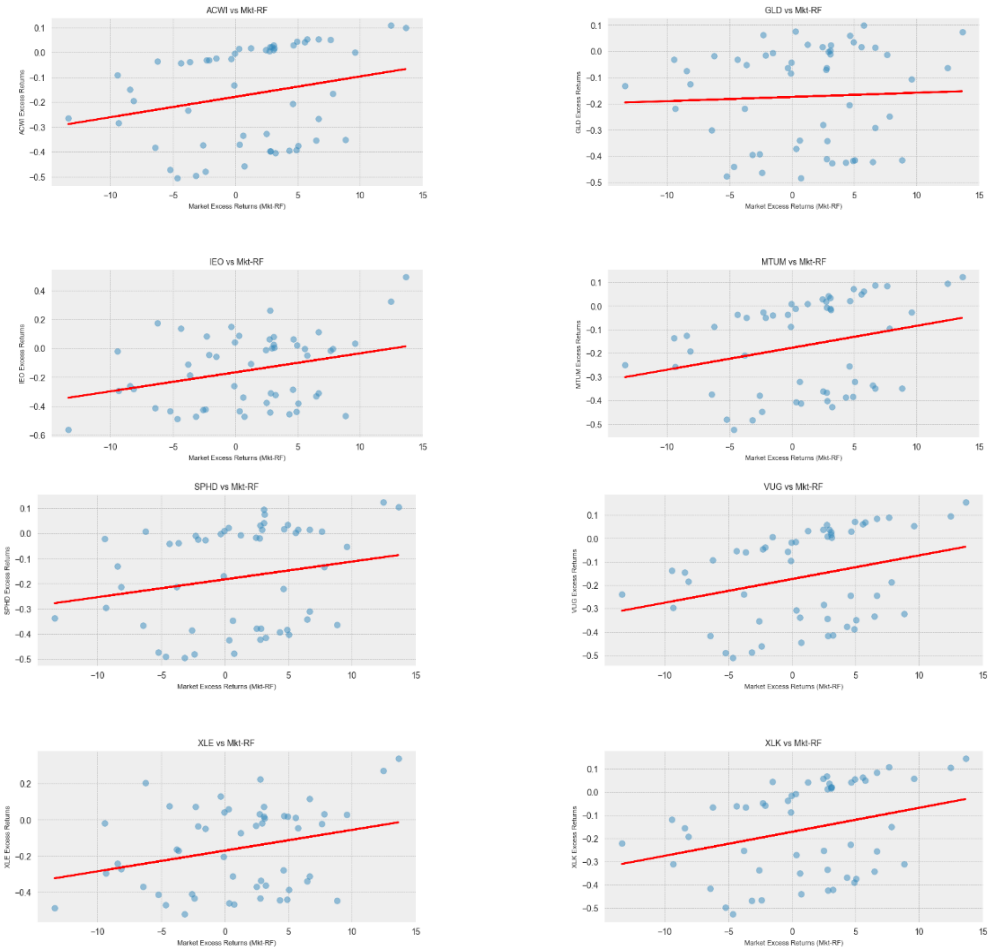



Figure 7: The plots display the relationship between the excess returns of various ETFs (such as ACWI, GLD, IEO, MTUM, SPHD, VUG, XLE, and XLK) and the market excess returns (Mkt-RF). Each plot shows a scatter of excess returns for the ETF compared to the market, with a regression line (in red) that represents the estimated relationship between the ETF and the market returns.

Finally, after calculating the factor loadings, we plot the regression results and adjust the portfolio weights accordingly (see Figure 7). By reducing exposure to unwanted factors such as SMB and increasing exposure to favorable factors such as HML, the portfolio is dynamically adjusted to reflect market conditions and investor strategy. The code performs regressions, calculates excess returns, and visualizes the relationship between ETF returns and market excess returns (Mkt-RF).

Across the plots, the regression lines have positive slopes, indicating that these ETFs tend to move in the same direction as the market (Mkt-RF). As the market excess return increases, the ETF's excess return generally increases too, suggesting a positive correlation with the market. Some ETFs, such as ACWI, GLD, and VUG, show a tighter clustering of points around the regression line, suggesting a more consistent relationship with the market returns. Other ETFs, such as IEO, XLE, and SPHD, show more scattered points, indicating more variability and a less consistent relationship with the market.

These visualizations in Figure 7 help in understanding the factor loadings of each ETF, particularly their sensitivity to the market factor (Mkt-RF). The plots help us to identify which ETFs may contribute more to market risk and which may provide a hedge or diversification benefit. Different ETFs exhibit varying degrees of market sensitivity, which can inform decisions about portfolio construction and factor tilting based on market conditions.

In the dynamic factor-based portfolio construction, we use the Fama-MacBeth regression method to adjust ETF weights based on empirical factor loadings. This approach provides a robust mechanism for dynamically managing factor exposures, optimizing the portfolio according to market conditions. We automate this adjustment by determining whether a factor loading is positive or negative and adjusting the weights accordingly. The goal is to either increase or decrease exposure to specific factors without hardcoding any adjustments, ensuring that portfolio constraints (such as maximum or minimum weight limits) are adhered to.

Three-Step Process for Dynamic Factor Adjustment:

A. Factor Loadings Assessment

Based on the factor loadings for each ETF, we determine whether to maximize or minimize exposure based on the following logic:

- Mkt-RF (Market Risk Premium): Maximize exposure if the loading is positive.
- SMB (Small Minus Big): Minimize if reducing small-cap exposure.
- HML (High Minus Low): Minimize for growth portfolios; maximize for value.
- RMW (Robust Minus Weak): Maximize exposure to profitable firms.
- CMA (Conservative Minus Aggressive): Maximize for conservative investment strategies.

B. Adjusting Weights: For each ETF, we adjust the portfolio weight based on its factor exposure:

- Increase the weight if a factor needs to be maximized.
- Decrease the weight if a factor needs to be minimized.

C. Normalization: After adjusting the weights, we normalize them to ensure they sum to 100%, maintaining the integrity of the overall portfolio allocation.

D. Threshold-Based Filtering of Factor Loadings: Next, we apply a thresholding mechanism to identify significant factor loadings. Using statistical thresholds (mean + standard deviation), we can isolate those loadings that meaningfully affect the portfolio’s exposure.

Factor Loadings for Each ETF:

| | Mkt-RF | SMB | HML | RMW | CMA |
|---|---|---|---|---|---|
| ACWI | 0.008 | 0.019 | -0.010 | 0.012 | 0.023 |
| GLD | 0.002 | 0.017 | -0.012 | 0.013 | 0.025 |
| IEO | 0.013 | 0.022 | 0.004 | 0.008 | 0.023 |
| MTUM | 0.009 | 0.017 | -0.011 | 0.009 | 0.024 |
| SPHD | 0.007 | 0.022 | -0.007 | 0.014 | 0.026 |
| VUG | 0.010 | 0.018 | -0.013 | 0.012 | 0.020 |
| XLE | 0.011 | 0.019 | 0.002 | 0.008 | 0.025 |
| XLK | 0.010 | 0.018 | -0.013 | 0.013 | 0.022 |

Calculated Threshold: 0.02085

Filtered Factor Loadings (Only those exceeding the threshold):

| | Mkt-RF | SMB | HML | RMW | CMA |
|---|---|---|---|---|---|
| **ACWI** | NaN | NaN | NaN | NaN | 0.023 |
| **GLD** | NaN | NaN | NaN | NaN | 0.025 |
| **IEO** | NaN | 0.022 | NaN | NaN | 0.023 |
| **MTUM** | NaN | NaN | NaN | NaN | 0.024 |
| **SPHD** | NaN | 0.022 | NaN | NaN | 0.026 |
| **VUG** | NaN | NaN | NaN | NaN | NaN |
| **XLE** | NaN | NaN | NaN | NaN | 0.025 |
| **XLK** | NaN | NaN | NaN | NaN | 0.022 |

Table 11: Initial and Filtered Factor Loadings

From the factor loadings, in an initial test, we apply a calculated threshold of 0.02085 that skewed the loadings towards positive values, eliminating nearly all negative loadings as shown above. This is clearly not favorable. Therefore there's a need to analyze the unique appearance of negative loadings and their values as follows:

Negative Loadings:

- -0.001: 1 time
- -0.007: 1 time
- -0.009: 3 times

A threshold of -0.008 strikes a balance between excluding insignificant values and retaining meaningful negative loadings.

Positive Loadings:

- 0.008 : 5 times
- 0.009 : 6 times
- 0.010 : 4 times
- 0.011 : 2 times
- 0.012 : 3 times
- 0.013 : 4 times
- 0.014 : 1 time

Similarly, positive loadings between 0.010 and 0.025 appear frequently, suggesting that 0.010 is an appropriate threshold for adjusting positive exposures. After using the new thresholds, the results is:

Filtered Factor Loadings (Only those exceeding the thresholds):

| | Mkt-RF | SMB | HML | RMW | CMA |
|---|---|---|---|---|---|
| **ACWI** | NaN | 0.019 | -0.010 | 0.012 | 0.023 |
| **GLD** | NaN | 0.017 | -0.012 | 0.013 | 0.025 |
| **IEO** | 0.013 | 0.022 | NaN | NaN | 0.023 |
| **MTUM** | NaN | 0.017 | -0.011 | NaN | 0.024 |
| **SPHD** | NaN | 0.022 | NaN | 0.014 | 0.026 |
| **VUG** | 0.010 | 0.018 | -0.013 | 0.012 | 0.020 |
| **XLE** | 0.011 | 0.019 | NaN | NaN | 0.025 |
| **XLK** | 0.010 | 0.018 | -0.013 | 0.013 | 0.022 |

The factor loadings for each ETF indicate how sensitive the ETF's returns are to each of the five Fama-French factors (Mkt-RF, SMB, HML, RMW, CMA). Here's what each of these loadings suggests about our W_a portfolio:

1. Mkt-RF (Market Risk Premium): A positive loading indicates that the ETF is sensitive to market movements. Most ETFs in this list show positive exposure, suggesting they generally move with the market.

2. SMB (Small Minus Big): Positive values suggest exposure to small-cap stocks, while negative values indicate large-cap exposure. For example, VB has a positive loading on SMB, indicating small-cap exposure.

3. HML (High Minus Low): Positive values suggest exposure to value stocks, while negative values indicate growth stock exposure. ETFs like SPHD and XLE show positive exposure, indicating a tilt towards value stocks.

4. RMW (Robust Minus Weak): This factor measures exposure to companies with robust profitability. Negative values, such as in XLE and IEO, suggest exposure to less profitable companies.

5. CMA (Conservative Minus Aggressive): Positive values indicate exposure to companies with conservative investment strategies, while negative values indicate aggressive strategies. For example, GLD shows a positive loading, indicating conservative exposure.

```
Starting Adjustments...

Adjustments for ACWI:
No significant adjustment for ACWI due to Mkt-RF factor.
Increasing exposure to ACWI due to positive SMB factor.
Decreasing exposure to ACWI due to negative HML factor.
Increasing exposure to ACWI due to positive RMW factor.
Increasing exposure to ACWI due to positive CMA factor.

Adjustments for GLD:
No significant adjustment for GLD due to Mkt-RF factor.
Increasing exposure to GLD due to positive SMB factor.
Decreasing exposure to GLD due to negative HML factor.
Increasing exposure to GLD due to positive RMW factor.
Increasing exposure to GLD due to positive CMA factor.

Adjustments for IEO:
Increasing exposure to IEO due to positive Mkt-RF factor
Increasing exposure to IEO due to positive SMB factor.
No significant adjustment for IEO due to HML factor.
No significant adjustment for IEO due to RMW factor.
Increasing exposure to IEO due to positive CMA factor.

Adjustments for MTUM:
No significant adjustment for MTUM due to Mkt-RF factor.
Increasing exposure to MTUM due to positive SMB factor.
Decreasing exposure to MTUM due to negative HML factor.
No significant adjustment for MTUM due to RMW factor.
Increasing exposure to MTUM due to positive CMA factor.

Adjustments for SPHD:
No significant adjustment for SPHD due to Mkt-RF factor.
Increasing exposure to SPHD due to positive SMB factor.
No significant adjustment for SPHD due to HML factor.
Increasing exposure to SPHD due to positive RMW factor.
Increasing exposure to SPHD due to positive CMA factor.

Adjustments for VUG:
Increasing exposure to VUG due to positive Mkt-RF factor
Increasing exposure to VUG due to positive SMB factor.
Decreasing exposure to VUG due to negative HML factor.
Increasing exposure to VUG due to positive RMW factor.
Increasing exposure to VUG due to positive CMA factor.

Adjustments for XLE:
Increasing exposure to XLE due to positive Mkt-RF factor
Increasing exposure to XLE due to positive SMB factor.
No significant adjustment for XLE due to HML factor.
No significant adjustment for XLE due to RMW factor.
Increasing exposure to XLE due to positive CMA factor.

Adjustments for XLK:
Increasing exposure to XLK due to positive Mkt-RF factor
Increasing exposure to XLK due to positive SMB factor.
Decreasing exposure to XLK due to negative HML factor.
Increasing exposure to XLK due to positive RMW factor.
Increasing exposure to XLK due to positive CMA factor.
```

Table 12: Left: Comparison of W_a and `adjusted_weights` which is the stored result after adjusting allocations based on factor loadings. Right: Performance of Portfolio metrics using the adjusted weights.

| | ETF | Previous Weight W_a | Adjusted Weight |
|---|---|---|---|
| 0 | VUG | 0.190 | 0.197 |
| 1 | SPHD | 0.080 | 0.084 |
| 2 | MTUM | 0.089 | 0.076 |
| 3 | ACWI | 0.100 | 0.095 |
| 4 | XLK | 0.171 | 0.178 |
| 5 | XLE | 0.125 | 0.131 |
| 6 | IEO | 0.076 | 0.080 |
| 7 | GLD | 0.169 | 0.160 |

Portfolio Metrics with Optimized Weights:

| | Metric | Value |
|---|---|---|
| 0 | Expected Return | 6560.95% |
| 1 | Volatility (Risk) | 89.88% |
| 2 | Sharpe Ratio | 72.94 |

This process ensures that significant factor exposures are captured while minimizing noise. By dynamically adjusting portfolio weights based on these filtered factor loadings, we can enhance portfolio performance while aligning with the investor's risk-return objectives.

### 4.6 Custom Factor BAB (Betting Against Beta)

In addition to the Fama-French factors, we introduce an interesting custom factor BAB (Betting Against Beta) which is a strategy that goes long on low-beta stocks and shorts high-beta stocks, betting that low-beta stocks will outperform on a risk-adjusted basis. We identify High-Beta and Low-Beta from the twenty (20) ETFs based on the betas calculated to classify these ETFs into high-beta (top 25%) and low-beta (bottom 25%) groups. Thereafter, we construct a BAB Portfolio that is long low-beta ETFs and short high-beta ETFs and compute the excess returns for the BAB portfolio over time and use it as a factor for backtesting.

Using a 60-day rolling window, we calculated the rolling beta and alpha of BAB portfolio to the SPY benchmark. The results were plotted to show BAB's risk exposure and performance fluctuated over time (see Figure 8). This analysis provided insights into the time-varying behavior of BAB factor, allowing us to better inform the portfolio optimization process. Additionally, we compared the average rolling beta and alpha of each factor to assess their historical contribution to risk-adjusted returns relative to the market.

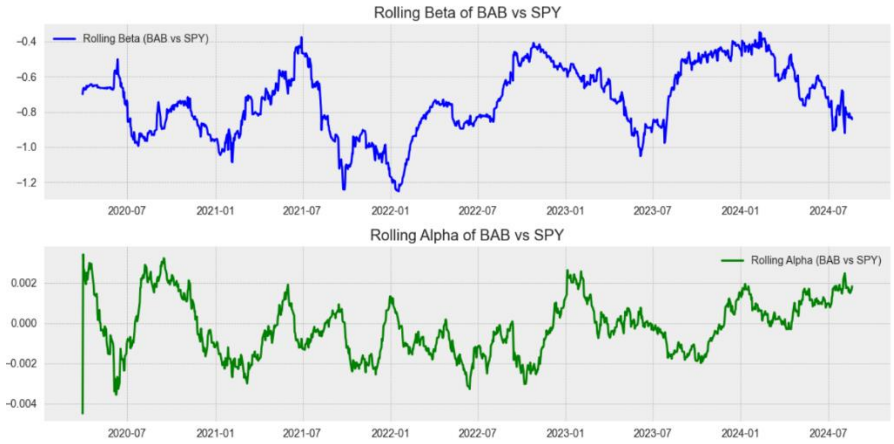



Figure 8: Plots of BAB Rolling Beta & Rolling Alpha vs SPY

Rolling Beta of BAB vs SPY:

1. The beta of the Betting Against Beta (BAB) strategy relative to SPY is consistently negative throughout the entire period, fluctuating between -1.4 and -2.2.

2. A negative beta indicates that the BAB strategy is inversely correlated with the market. When the market (SPY) rises, the BAB strategy tends to decline, and vice versa.

3. Notably, the beta becomes more negative during certain periods (e.g., late 2020 and early 2024), indicating that the BAB strategy's inverse relationship with the market has strengthened during these times.

4. In mid-2024, there is a sharp downward spike in beta, potentially signifying a period of high market volatility or a significant shift in how high- and low-beta assets are behaving.

Rolling Alpha of BAB vs SPY:

1. The alpha of the BAB strategy fluctuates around zero for most of the period, with both positive and negative phases. However, the alpha remains quite low throughout, suggesting that the BAB strategy does not consistently generate strong excess returns.

2. Alpha represents the excess return that the BAB strategy generates over what would be expected based on its beta (market exposure). Positive alpha implies outperformance relative to the market.

3. From early 2021 until mid-2022, the alpha stays slightly negative, suggesting underperformance compared to expectations.

4. Around mid-2023, the rolling alpha shows signs of improvement, and there is a noticeable upward trend, indicating that the BAB strategy started to produce positive excess returns relative to SPY.

We generated the performance metrics of the BAB (Betting Against Beta) strategy which shows a significant underperformance compared to the SPY (S&P 500) benchmark in this analysis:

```
BAB Annual Return: -23.54%, Volatility: 31.40%, Sharpe Ratio: -0.91
SPY Annual Return: 15.54%, Volatility: 21.52%, Sharpe Ratio: 0.49
```

Performance Metrics for BAB vs SPY:

1. Negative annual return of the BAB strategy at -18.64% indicates poor performance during the analysis period. Given that the BAB strategy is built on long low-beta and short high-beta positions, the negative performance likely reflects a period where high-beta stocks outperformed low-beta stocks, which would hurt the BAB portfolio.

2. The BAB strategy is highly risky and volatile, with volatility higher than SPY.

3. The negative Sharpe ratio confirms that the BAB strategy delivered negative risk-adjusted returns.

4. SPY delivered a solid positive return, positive Sharpe ratio, and lower volatility compared to BAB, which overall indicating a strong bull market during the backtesting period from 2020-2024.

Insights on BAB Strategy Performance with ETFs:

1. Market-Cap Weighting in ETFs: One of the primary issues with applying the BAB (Betting Against Beta) strategy to ETFs is that ETFs are often market-cap weighted. This means that larger, more liquid stocks dominate the portfolio, which dilutes the BAB strategy's effectiveness. BAB relies on long positions in low-beta stocks and short positions in high-beta stocks, but the composition of ETFs typically underrepresents high-beta, small-cap stocks, reducing the potential impact of the BAB factor.

2. Mismatch in Targeted Stock Categories: Research shows that BAB's success comes from targeting smaller, more volatile high-beta stocks, which tend to underperform on a risk-adjusted basis. However, because ETFs focus primarily on large-cap stocks, the effectiveness of the BAB strategy is diminished. The price movements of these small, high-beta stocks, which drive BAB returns, are not adequately captured by large-cap-dominated ETFs.

3. Bull Market Impact: The backtesting period from 2020-2024 coincided with a bull market. High-beta stocks typically perform well during such periods, which undermines the short side of the BAB strategy. As a result, BAB strategies that bet against high-beta stocks would have been unprofitable during this time, contributing to the poor performance.

4. Suboptimal Market Environment for BAB: The poor absolute and risk-adjusted returns of the BAB strategy reflect an unfavorable environment for low-beta strategies. Investors using the BAB approach should reconsider the timing of such a strategy, particularly in a bull market, or explore other factor-based approaches that better align with current market conditions. Given the strong performance of SPY, which benefits from high-beta and broad market exposure, alternative strategies focused on high-beta or large-cap stocks may yield better results in similar environments.

### 4.7 Macroeconomic Factors - Global Growth, Global Inflation, Global Liquidity and Risk Appetite

The eight (8) optimal assets we have selected act as proxies for U.S. and industry-specific factors. In this section we complement the selection with macroeconomic analysis. In order to consider the macroeconomic situation into investor views, we chart key four (4) macroeconomic factors — Global Growth, Global Inflation, Global Liquidity, and Risk Appetite — over the period from 1996 to 2023. These four factors are based on Fullerton's framework for investment returns (Clair, 2022), and we replicate them using available public and proxy data.

**Integration of Macro Indicators into Black-Litterman Model:** in addition to traditional inputs like equilibrium returns and investor views, the Black-Litterman model can be enhanced by incorporating macroeconomic indicators, which provide insights into broader market trends and conditions. Specifically, the following four indicators can be integrated as "views" into the model which asset management firm Fullerton uses as part of their proprietary framework and methodology, utilized to generate insights into market dynamics and asset allocation decisions:

1. Global Growth: This indicator, derived from a combination of major global indices such as the S&P 500, Nikkei 225, and Euro Stoxx 50, provides a normalized measure of economic growth across key regions. It captures the overall economic momentum, which can influence expected returns:

```
# Define the tickers for global growth
growth_indices = {
    'S&P 500': '^GSPC',
    'Nikkei 225': '^N225',
    'Hang Seng': '^HSI',
    'Shanghai Composite': '000001.SS',
    'Euro Stoxx 50': '^STOXX50E',
    'Nifty 50': '^NSEI'
}
```

2. Global Inflation: Utilizing Gold (GLD) prices as a proxy. Inflation expectations can impact asset prices and returns, as higher inflation often leads to higher interest rates, affecting bond yields and equity valuations.

3. Global Liquidity: Measured by indicators such as the 10-Year Treasury Yield (TNX). High liquidity typically supports asset prices, while tightening liquidity conditions can lead to increased volatility and lower valuations.

4. Risk Appetite: Using the VIX index as a proxy, this indicator gauges market sentiment and risk tolerance. A higher VIX indicates greater market volatility and risk aversion, which can lead to changes in asset allocation preferences.

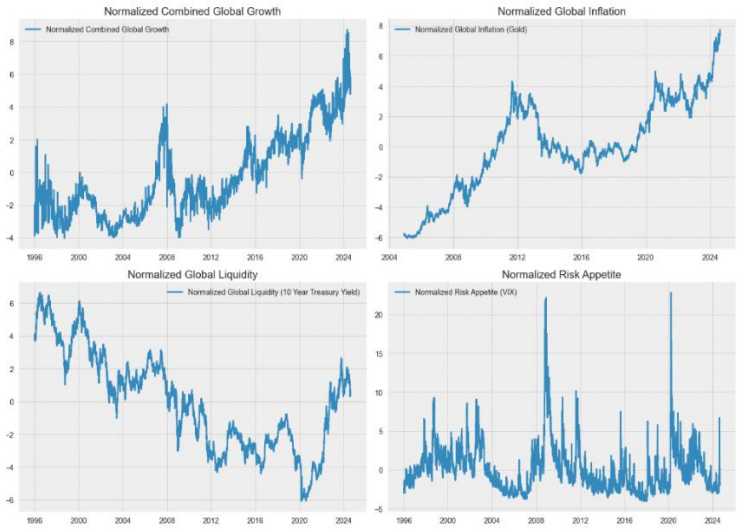



Figure 9: Time series plots for the four Macroeconomic factors using data from January 1996 to August 2024.

Not satisfied with the time series plots in Figure 9 above, we adjusted the sources of data by replacing GLD and TNX proxy data with macroeconomic data sources from the World Bank and BIS to generate Figure 10 of improved plots:

1. Global Inflation: Utilizing inflation data from the World Bank's Prospects Group represented by metrics like the Median, Average, and GDP-weighted Average inflation rates, offers insights into the inflationary environment.

Data Collection for Global Inflation (Alternative to using Gold): The World Bank's Prospects Group has constructed a global database of inflation dated April 2024 version. The database covers up to 209 countries over the period 1970-2023 and includes six measures of inflation in three frequencies (annual, quarterly, and monthly) for Headline consumer price index (CPI) inflation.

We extract the Global headline consumer price index (HCPI) inflation data from the excel file downloaded from World Bank's website[9] as an alternative to using Gold prices in our previous step. This field provides a comprehensive measure of inflation by tracking the changes in the price level of a basket of consumer goods and services across the globe, making it an ideal indicator to plot global inflation trends. This is shown in code:

```
# Transpose the data for easier manipulation
inflation_data_transposed = inflation_data_agg.transpose()

# Set the first row as the header after transposing
new_header = inflation_data_transposed.iloc[0]
inflation_data_transposed = inflation_data_transposed[1:]
inflation_data_transposed.columns = new_header

# Remove the non-year rows
inflation_data_transposed = inflation_data_transposed.drop(['Description', 'Aggregation', 'Sample country'], axis=0)

# Parse the years as integers and reset index
inflation_data_transposed.index = pd.to_numeric(inflation_data_transposed.index).astype(int)
inflation_data_transposed = inflation_data_transposed.reset_index()

# Dropping HCPI_GLOBAL_MED_BAL ('Median (balanced group)') for clearer plot display
inflation_data_transposed = inflation_data_transposed.drop('HCPI_GLOBAL_MED_BAL', axis=1)

inflation_data_transposed.tail(5)
```

| Series Name | index | HCPI_GLOBAL_MED | HCPI_GLOBAL_AVE | HCPI_GLOBAL_AVE_GDP |
|---|---|---|---|---|
| 49 | 2019 | 2.204 | 3.144 | 2.432 |
| 50 | 2020 | 2.255 | 3.831 | 2.117 |
| 51 | 2021 | 3.356 | 5.056 | 3.735 |
| 52 | 2022 | 8.082 | 9.575 | 6.704 |
| 53 | 2023 | 5.871 | 7.859 | 4.633 |

2. Global Liquidity: Utilizing data from the Bank for International Settlements (BIS) global liquidity indicators (GLIs), which reflects the availability of liquidity in the financial system.

Data Collection for Global Liquidity (Alternative to using 10Y Treasury Yield): To create the Global Liquidity plot, we used data from the Bank for International Settlements (BIS) global liquidity indicators (GLIs), specifically the file[10] "WS_GLI_csv_col.csv." , instead of using the 10 Year Treasury Yield as proxy. The BIS defines "global liquidity" as the ease of financing in global financial markets. The GLIs track credit to non-bank borrowers, covering both bank loans and funding from global bond markets through international debt securities (IDS). The focus is on foreign currency credit denominated in major reserve currencies (US dollars, euros, Japanese yen) to non-residents.

The code below filters the excel data to isolate specific conditions (i.e., All currencies, all countries, all sectors, banks total lending), normalizes the selected data, and then visualizes it as a bar chart to represent normalized global liquidity trends:

```
# Create a condition that checks data pulling criteria: All currencies, all countries, all sectors, Banks-total.
condition = (data['L_POS_TYPE'] == 'I') & (data['BORROWERS_CTY'] == '5J') & (data['BORROWERS_SECTOR'] == 'A')
          & (data['UNIT_MEASURE'] == '770')

# Use the condition to filter the DataFrame
global_liquidity_data = data[condition]
global_liquidity_data
```

| | FREQ | Frequency | CURR_DENOM | Currency of denomination | BORROWERS_CTY | Borrowers' country | BORROWERS_SECTOR | Borrowers' sector | LENDERS_SECTOR | Lending sector | ... | 2( |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 142 | Q | Quarterly | TO1 | All currencies | 5J | All countries (total) | A | All sectors | B | Banks, total | ... | 45. |

1 rows × 209 columns

[9] https://www.worldbank.org/en/research/brief/inflation-database

[10] https://data.bis.org/topics/GLI

**Analysis of the Four Global Macroeconomic Factors (see Figure 10 – improved plots):**

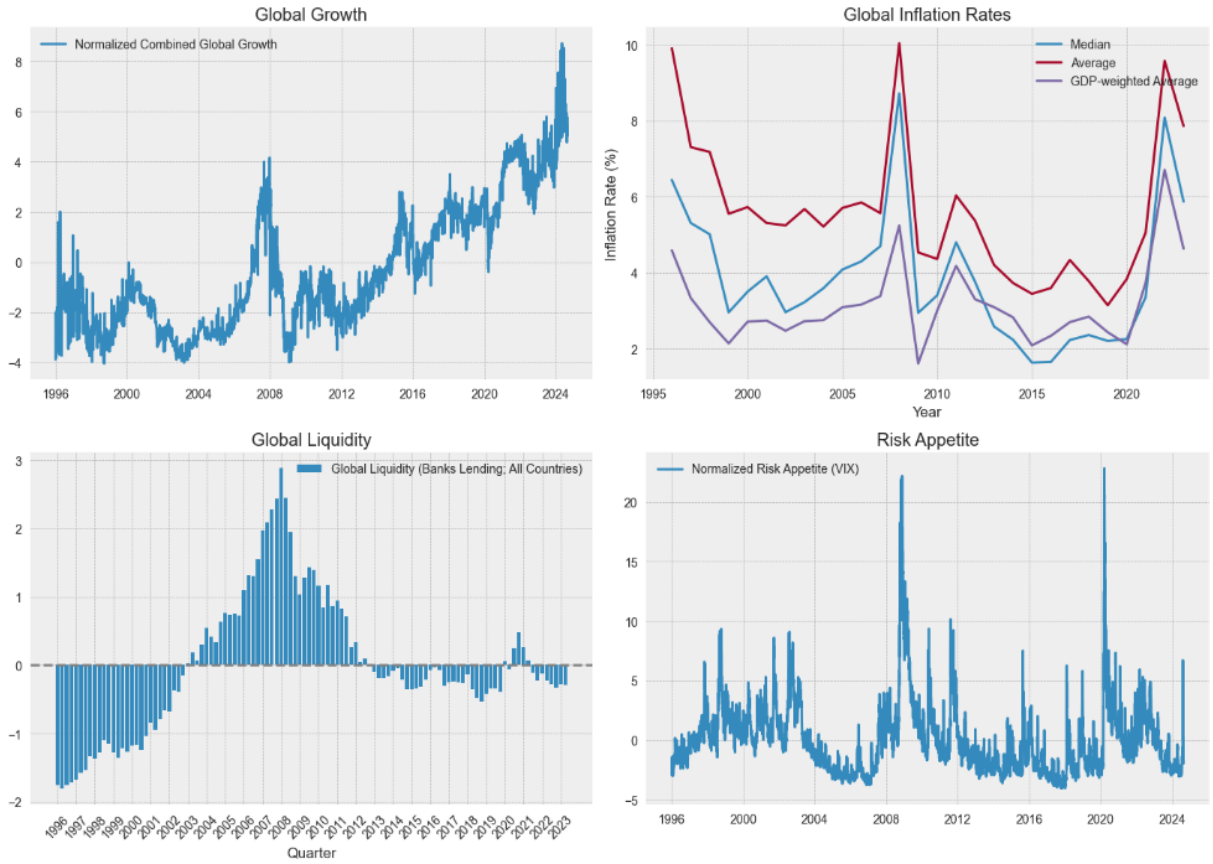



Figure 10: Global Growth, Global Inflation, Global Liquidity and Risk Appetite from January 1996 to August 2024.

1. Global Growth: The plot shows the normalized trend of economic growth using indices like the S&P 500, Nikkei 225, and Euro Stoxx 50. It highlights the overall economic momentum and reflects the impact of global economic cycles, such as the dot-com bubble, the 2008 financial crisis, and recent recovery phases. Since 2020, the plot shows significant growth momentum, especially post-COVID-19, where the global economy experienced a sharp rebound as countries reopened and economic activities resumed. The normalization of growth indices highlights strong recoveries, particularly in sectors like technology and consumer goods. This trend is evident in the marked rise in the growth index, surpassing pre-pandemic levels.

2. Global Inflation: Inflationary pressures have been moderating recently from the higher inflation post-pandemic due to supply chain disruptions and increased demand. This plot, based on data from the World Bank's Prospects Group, illustrates global inflation trends using Median, Average, and GDP-weighted Average rates. The data spans from 1995 to 2023, capturing fluctuations in inflation rates. The plot indicates varying inflation rates, with a noticeable spike in recent years. Here are certain geopolitical considerations that will affect inflation rates going forward:

    a. If Donald Trump is re-elected as President of the United States in 2024, there are expectations of increased inflation. Trump's economic policies may expand significant tariffs on Chinese goods and potentially other imports, which could raise consumer prices.

    b. Japan's Currency Policies: Japan has been actively managing its currency, the yen, in response to global economic conditions. Recently, the yen has seen fluctuations due to differing monetary policies between

Japan and other major economies like the United States. Japan's efforts to keep interest rates low and maintain a relatively weak yen aim to support exports and stimulate economic growth.

c. China's economy continues to face challenges, partly due to the ongoing trade war with the United States. The trade tensions, marked by tariffs and technological restrictions, have strained China's economic growth. The country has seen reduced export opportunities and increased costs for imported goods, impacting industrial production and overall economic activity. Additionally, internal policies and external pressures, such as those from the U.S. and its allies restricting technology exports, have further complicated China's economic landscape

3. Global Liquidity: The plot uses data from the BIS global liquidity indicators, focusing on bank loans and international debt securities. It tracks the ease of financing in global markets, with periods of high liquidity supporting asset prices and low liquidity potentially leading to increased volatility. The data highlights a notable increase in liquidity during COVID-19 and a liquidity crunch post-COVID exacerbated by high interest rates. However, recent data shows that liquidity conditions are improving.

4. Risk Appetite: The risk appetite situation is staying positive. This plot helps understand shifts in investor behavior, particularly during market stress events such as the 2008 crisis and the COVID-19 pandemic. But post-COVID, there has been a reduction in volatility as markets stabilized, but occasional spikes indicate ongoing uncertainties with SPY soaring to new highs in recent months fueled by the AI boom.

The above macroeconomic perspective informs our weight allocation, ensuring that investor views are aligned with broader economic conditions. Based on the analysis, we implement the Black-Litterman model that adjusts returns based on static views and fixed confidence levels. For example, if global liquidity is tightening, one could lower the model's confidence in high-growth sectors and increase confidence in defensive assets, and this will affect the omega values.

### 4.8 Rationale on Investor's Views (as of August 21, 2024)

The views matrix Q indicate the investor's subjective views on the expected excess returns of specific assets or groups of assets:

- View 1: expects GLD to outperform by 15% relative to MTUM.
- View 2: expects XLK to outperform by 8% relative to XLE.
- View 3: expects SPHD to outperform by 6% relative to VUG.

```
investor_views = [
    ('GLD', '>', 'MTUM', 0.15),
    ('XLK', '>', 'XLE', 0.08),
    ('SPHD', '>', 'VUG', 0.06)
]
```

View 1. Gold (GLD) vs. Momentum (MTUM) Stocks: Given the high inflation environment, economic uncertainties, and the backdrop in escalation of war in the Middle East, gold is expected to outperform momentum stocks as a hedge against high inflation. On the 3rd of August 2024, Iran and their allied groups vow to hit back at Israel in response to the killing of Hamas leader by Israel, while Washington increases military presence in the Middle East.

View 2. Technology (XLK) vs. Energy (XLE): Despite experiencing high volatility, the Technology sector is anticipated to outperform the Energy sector. This expectation is driven by strong underlying fundamentals and continuous innovation, even as commodity prices may stabilize and the initial excitement around AI cools. The long-term potential of Generative AI remains promising, and Technology companies continue to report robust earnings. However, during mid-August 2024 the investors were asking about their AI investments to 'show me the money'. Nevertheless, we maintain a positive outlook in the Technology sector.

View 3. High Dividend (SPHD) vs. Growth Stocks (VUG): With a high-interest rate environment, high dividend stocks may offer a more stable income stream compared to growth stocks, of which growth stocks could be more sensitive to rising rates, and the escalation of war in the Middle East. U.S. stocks are expected to gradually increase as cost pressures stabilize and corporate earnings start to recover. We are seeing a recovering trend particularly in Asia

and Europe. In the U.S., a positive surprise in earnings could propel the market further. Although U.S. inflation is decreasing at a slower pace than anticipated, the general expectation remains that central banks in developed markets will begin lowering interest rates later this year.

**Integrating Market Equilibrium and Investor Views with Uncertainty and Scaling**

To incorporate investor perspectives into the Black-Litterman (BL) model for the eight (8) selected ETFs, we integrate macroeconomic factors such as inflation expectations, liquidity conditions, and the broader economic outlook. This integration enables a customized portfolio optimization process.

- Q Matrix (Views Vector): This matrix quantifies the investor's specific views on the relative performance of assets. Each element in Q represents the expected outperformance of one asset over another in a given view. For example, if an investor expects ETF1 to outperform ETF2, this will be reflected in the corresponding Q element.

Views Matrix (Q):

| | Views |
|---|---|
| 0 | 0.150 |
| 1 | 0.080 |
| 2 | 0.060 |

- P Matrix (Link Matrix): This matrix links each investor's view to the relevant assets. For a view where ETF1 is expected to outperform ETF2, +1 is assigned to ETF1's column and -1 to ETF2's column. This approach ensures the model understands which assets are involved in each view and the direction of expected outperformance or underperformance.

Link Matrix (P):

| VUG | SPHD | MTUM | ACWI | XLK | XLE | IEO | GLD |
|---|---|---|---|---|---|---|---|
| 0 | 0 | -1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | -1 | 0 | 0 |
| -1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |

After constructing the P and Q matrices, the Black-Litterman model combines these investor views with the market's implied equilibrium returns (Π) and the covariance matrix of asset returns. The Black-Litterman model combines the implied market equilibrium returns (Π) with the adjusted views (through Q and P), weighted by the uncertainty in the views (Ω) and the scalar (τ). Earlier in computing Π, we use the risk aversion coefficient (λ) of the U.S. market at roughly 3.07, taken from the work of Idzorek (2007). This combination creates the new expected return vector called the BL_return_vector in the code, which reflects both market equilibrium and subjective views. The formula is stated as:

$$\text{BL_return_vector} = \Pi + \sigma_{scaled} \cdot P^T \cdot (P \cdot \sigma_{scaled} \cdot P^T + \omega)^{-1} \cdot (Q - P \cdot \Pi)$$

The scalar (τ) and uncertainty matrix (Ω) determine the relative influence of the implied equilibrium returns and the investor's views on the final adjusted returns. A higher τ or lower Ω value gives more weight to the views, while a lower τ or higher Ω value places more emphasis on the implied equilibrium returns.

In the BL model, the covariance matrix is used to minimize the portfolio variance for a given level of expected return N×N matrix, where N is the number of assets or factors in the portfolio. This is at the heart of mean-variance optimization. In our case, N=8 corresponding to the 8 ETFs 'optimal bets' defined in the investment universe.

The Link Matrix is a K×N matrix, where K is the number of views, and N is the number of assets or factors. While the Views Vector is Kx1 vector containing the expected returns specified by investor's views.

**Code Implementation:**

The following code implements the Black-Litterman model to adjust the returns and covariance matrix based on investor views and then performs portfolio optimization. The confidence level of the views shall be set from 70-90%.

We define omega as a diagonal matrix representing a specific confidence level from 70%-90% for each view, instead of a scalar matrix applying 70% confidence to all views. If there are multiple views, omega should have a value corresponding to the confidence for each view on its diagonal.

The code is shown here and the covariance matrix ('C'):

```
# Confidence levels for each view (0 < confidence <= 1)
confidence_levels = [0.90, # escalation of war in the Middle East
                     0.70, # positive outlook on tech compared to energy
                     0.80] # high interest rate to continue and growth stocks are sensitive to it

# Calculate the uncertainty matrix omega (inverse of confidence levels)
omega = np.diag([1 / c for c in confidence_levels])

# Scaling factor (tau): measure of uncertainty of the prior estimate of the mean returns
tau = 0.1
sigma_scaled = tau * C
```

```
# Check on the covariance matrix
C
```

| | VUG | SPHD | MTUM | ACWI | XLK | XLE | IEO | GLD |
|---|---|---|---|---|---|---|---|---|
| VUG | 0.028 | 0.027 | 0.017 | 0.020 | -0.019 | 0.033 | 0.022 | -0.007 |
| SPHD | 0.027 | 0.034 | 0.016 | 0.022 | -0.033 | 0.044 | 0.030 | -0.008 |
| MTUM | 0.017 | 0.016 | 0.012 | 0.013 | -0.008 | 0.021 | 0.015 | -0.004 |
| ACWI | 0.020 | 0.022 | 0.013 | 0.016 | -0.017 | 0.028 | 0.020 | -0.005 |
| XLK | -0.019 | -0.033 | -0.008 | -0.017 | 0.099 | -0.038 | -0.014 | 0.016 |
| XLE | 0.033 | 0.044 | 0.021 | 0.028 | -0.038 | 0.074 | 0.058 | -0.010 |
| IEO | 0.022 | 0.030 | 0.015 | 0.020 | -0.014 | 0.058 | 0.053 | -0.005 |
| GLD | -0.007 | -0.008 | -0.004 | -0.005 | 0.016 | -0.010 | -0.005 | 0.005 |

Thereafter, we adjust the returns (‘adjusted_returns’) in the Black-Litterman model to incorporate investor views into the expected returns of the portfolio. The standard market-implied returns (Π) are based on market equilibrium, but investor views provide additional insights that may differ from the market consensus.

```
# Black-Litterman formula to adjust returns

# Step 1: Calculate P @ sigma_scaled @ P.T
P_sigma_Pt = P @ sigma_scaled @ P.T
print("Shape of P @ sigma_scaled @ P.T:", P_sigma_Pt.shape)

# Step 2: Add omega to the result and invert
inv_term = np.linalg.inv(P_sigma_Pt + omega)
print("Shape of inv(P @ sigma_scaled @ P.T + omega):", inv_term.shape)

# Step 3: Calculate Q - P @ Pi
Q_minus_Pi = Q - P @ Pi
print("Shape of Q - P @ Pi:", Q_minus_Pi.shape)

# Step 4: Calculate sigma_scaled @ P.T @ inv_term
sigma_Pt_inv = sigma_scaled @ P.T @ inv_term
print("Shape of sigma_scaled @ P.T @ inv_term:", sigma_Pt_inv.shape)

# Step 5: Final step: sigma_Pt_inv @ (Q - P @ Pi)
final_term = sigma_Pt_inv @ Q_minus_Pi
print("Shape of final term (sigma_scaled @ P.T @ inv_term @ (Q - P @ Pi)):", final_term.shape)

# Step 6: Calculate the adjusted returns
adjusted_returns = Pi + final_term
print("Shape of adjusted returns:", adjusted_returns.shape)

print("Adjusted Returns:\n", adjusted_returns)
```

```
Shape of P @ sigma_scaled @ P.T: (3, 3)
Shape of inv(P @ sigma_scaled @ P.T + omega): (3, 3)
Shape of Q - P @ Pi: (3,)
Shape of sigma_scaled @ P.T @ inv_term: (8, 3)
Shape of final term (sigma_scaled @ P.T @ inv_term @ (Q - P @ Pi)): (8,)
Shape of adjusted returns: (8,)
Adjusted Returns:
 VUG     0.038
SPHD     0.036
MTUM     0.026
ACWI     0.029
XLK      0.017
XLE      0.060
IEO      0.054
GLD     -0.002
dtype: float64
```

By adjusting the returns (results shown above), we blend the market equilibrium returns with the investor's specific expectations, while also accounting for the confidence in those views. This adjustment ensures the new expected returns reflect both the broad market outlook and the investor’s perspective, resulting in a more personalized and informed return forecast for portfolio optimization.

In a similar way, we adjust the covariance matrix (‘adjusted_covariance’) in the Black-Litterman model as well to reflect how investor views affect the portfolio's risk structure. Investor views, such as expecting one asset to outperform another, can change the relationships between assets (correlations and volatilities).

```
Adjusted Covariance Matrix:
```

| | VUG | SPHD | MTUM | ACWI | XLK | XLE | IEO | GLD |
|---|---|---|---|---|---|---|---|---|
| VUG | 0.030000 | 0.029000 | 0.019000 | 0.022000 | -0.020000 | 0.037000 | 0.025000 | -0.007000 |
| SPHD | 0.029000 | 0.038000 | 0.018000 | 0.024000 | -0.037000 | 0.048000 | 0.033000 | -0.009000 |
| MTUM | 0.019000 | 0.018000 | 0.013000 | 0.014000 | -0.009000 | 0.023000 | 0.017000 | -0.004000 |
| ACWI | 0.022000 | 0.024000 | 0.014000 | 0.018000 | -0.019000 | 0.031000 | 0.021000 | -0.006000 |
| XLK | -0.020000 | -0.037000 | -0.009000 | -0.019000 | 0.109000 | -0.042000 | -0.016000 | 0.018000 |
| XLE | 0.037000 | 0.048000 | 0.023000 | 0.031000 | -0.042000 | 0.081000 | 0.064000 | -0.011000 |
| IEO | 0.025000 | 0.033000 | 0.017000 | 0.021000 | -0.016000 | 0.064000 | 0.058000 | -0.006000 |
| GLD | -0.007000 | -0.009000 | -0.004000 | -0.006000 | 0.018000 | -0.011000 | -0.006000 | 0.005000 |

The adjustment accounts for both the confidence in these views and their impact on overall risk. This ensures that the portfolio's risk profile aligns with the new expected returns, integrating both market equilibrium and investor expectations in a consistent way.

**Calculating the New Combined Return Vector:** In the Black-Litterman model, the New Combined Return Vector (often denoted as $E[R]$) is the result of combining the market equilibrium returns (Π) with the investor's views. This adjusted return vector $\mu_{implied}$ or $E[R]$ is what the model recommends as the expected returns after incorporating both market consensus and the investor's specific views.

Now, we use the formula to calculate the adjusted expected returns $\mu_{implied}$ that reflect both the market consensus and the investor's views, weighted by confidence levels:

$$\mu_{implied} = \Pi + \tau \Sigma \cdot P^T \cdot (P \cdot \tau \Sigma \cdot P^T + \Omega)^{-1} (Q - P \cdot \Pi)$$

Where:

- Π was earlier calculated using the formula: Π = risk_aversion × np.dot ($C$,$Wa$), and using the adjusted weights $Wa$ rather than the raw market capitalization weights $W$.

- For the value of Tau (τ), we adhere to the precedent established by Pyo & Lee (2018) and set it to 0.1 accordingly.

- $\mu_{implied}$ or $E[R]$: Π directly influences the weight allocation, reflecting the market equilibrium adjusted by the investor's views. Best used when the goal is to determine how the market's implied returns should theoretically influence portfolio allocation, assuming the investor's views are fully incorporated.

The table below is generated for comparison and analysis that is similar to Table 6 from the "Guide to Black-Litterman Model" by Thomas Idzorek.

Table 13: Return Vectors and Resulting Portfolio Weights for Selected ETFs under Black-Litterman Adjustments

| Asset Class | New Combined Return Vector E[R] | Implied Equilibrium Return Vector Π | Difference E[R] – Π | New Weight ŵ | Market Capitalization Weight w_mkt | Difference ŵ – w_mkt |
|---|---|---|---|---|---|---|
| VUG | 0.037 | 0.038 | -0.001 | 14.45% | 22.84% | -8.40% |
| SPHD | 0.035 | 0.036 | -0.001 | 13.79% | 8.05% | 5.73% |
| MTUM | 0.026 | 0.026 | -0.001 | 10.05% | 8.78% | 1.27% |
| ACWI | 0.028 | 0.029 | -0.001 | 10.97% | 9.73% | 1.24% |
| XLK | 0.018 | 0.017 | 0.002 | 7.12% | 15.59% | -8.47% |
| XLE | 0.060 | 0.061 | -0.001 | 23.23% | 11.92% | 11.32% |
| IEO | 0.054 | 0.055 | -0.001 | 21.17% | 7.74% | 13.42% |
| GLD | -0.002 | -0.002 | 0.000 | -0.77% | 15.35% | -16.12% |

Even though the expressed views may directly involve some ETFs, all assets' expected returns change slightly from their implied equilibrium returns, as evidenced by the small differences in column 3 (E[R] − Π). This is consistent with the Black-Litterman framework, where the covariance matrix (Σ) ensures that all asset returns are interconnected.

- VUG: As expected, VUG is reduced in weight (-8.40%), aligning with the View 3 that growth stocks will be more sensitive to rising interest rates.

- SPHD: Expected to outperform VUG by 6% with a confidence level of 0.80, reflecting investor View 3 that high interest rates will continue, benefiting dividend-focused, low-volatility stocks. This view is clearly reflected in SPHD's weight that increases by 5.73%, signaling an overweight in this ETF relative to its market capitalization weight.

- XLK and XLE: Despite View 2 expecting XLK to outperform XLE, the broader market or macro factors result in XLE being more heavily weighted.

- XLE and IEO: The positive adjustments (though small) in energy-related sectors are reflected in significant increases in portfolio weights: XLE by 11.32% and IEO by 13.42%. This could indicate that the views on these sectors (despite minor changes in expected returns) align with investor optimism, likely driven by macroeconomic factors like rising energy prices.

- GLD: Gold sees no change in its expected return (-0.002 remains unchanged), yet its weight dramatically decreases by 16.12%. Despite the expectation of GLD outperforming, the new weight of -0.77% for GLD is significantly lower than its market cap weight of 15.35%, indicating an underweight position. This might be due to the combined impact of other views or a balancing effect from the covariance matrix.

**Summary of Analysis on the Broader Portfolio Adjustments:** The portfolio reflects a tilt towards defensive assets and energy sectors (with large allocations to XLE and IEO), despite the positive view on technology (XLK). This suggests that investor sentiment is cautious, hedging against potential volatility due to geopolitical risk (confidence level of 90% for GLD) and economic uncertainty (with interest rates likely to remain high). Energy ETFs (XLE, IEO) still hold strong positions, even with a tech-favorable outlook (XLK). SPHD is the clearest beneficiary of the current interest rate environment, while VUG and growth-focused assets are scaled back in response to macro concerns.

### 4.9 Portfolio Optimization

Drawing on the work of Malkiel and Fama in "Efficient Capital Markets: A Review of Theory and Empirical Work" (1970), we recognize the challenge of predicting asset returns, which attributed to two (2) elements: the semi-strong form of market efficiency, which implies that all publicly available information is reflected in stock prices, and the significant noise inherent in financial data. These challenges make portfolio optimization a complex task and difficult for investors to consistently achieve above-market returns, further highlighting the relevance of the Black-Litterman model in incorporating investor views and equilibrium market conditions into portfolio construction.

Before engaging in portfolio optimization, systematic backtesting was conducted to evaluate each ETF's exposure to market risk and its potential for generating excess returns. Using a 60-day rolling window, we calculated the rolling beta and alpha of each ETF relative to the SPY benchmark. The results were plotted to show how each ETF's risk exposure and performance fluctuated over time (see Section 4.2 Figures 2 & 3).

The portfolio weights are computed using four (4) optimization techniques: mean-variance optimization, Maximum Sharpe Ratio, higher-order moments, and Minimum Tracking Error of which we use SPY as the benchmark index.

To determine the optimal portfolio allocations, we rely on quadratic optimization routines in some of the portfolio optimization process, which are particularly well-suited for solving the constrained portfolio optimization problem inherent in the Black-Litterman model. Quadratic optimization minimizes the portfolio's risk (variance) while considering the expected returns and the covariance of the assets.

This optimization problem is defined as:

$$\min_{w} \frac{1}{2} w^T \sum w - \mu^T w$$

Where:

- $w$ is the vector of portfolio weights.
- $\Sigma$ is the covariance matrix of asset returns.
- $\mu$ is the vector of expected returns (adjusted for investor views in the Black-Litterman model).

The two constraints in this optimization are:

1. $\sum w_i = 1$, ensuring that the total portfolio weights sum to 1 (i.e., fully invested).
2. $w_i \geq 0$, to prevent short-selling.

By using quadratic programming, we ensure that the final portfolio maximizes expected return for a given level of risk, providing an efficient and accurate method for optimizing asset allocations. The codes in sections of portfolio optimization performs this quadratic optimization using 'scipy.optimize.minimize' to find the portfolio weights that achieve the desired trade-off between risk and return, ensuring the portfolio is well-diversified and aligned with the investor's risk tolerance. The method used 'scipy.optimize.minimize' function is SLSQP (Sequential Least Squares Programming) that is suitable for constrained optimization problems.

**Mean-Variance Optimization:** Mean-Variance Optimization is a fundamental concept in modern portfolio theory, introduced by Harry Markowitz in 1952. It provides a mathematical framework for constructing portfolios that maximize expected returns for a given level of risk (variance) or minimize risk for a given expected return. This method is widely used in portfolio management to achieve the most efficient allocation of assets, balancing risk, and reward.

We use Python's scipy.optimize.minimize function, adjusted weights that was previously computed ($W_a$) and covariance matrix (C) to find the optimal portfolio weights that minimize the portfolio's variance (risk) based on a given covariance matrix of asset returns. The result is optimal asset allocation that minimizes portfolio risk given the assets' covariance:

```
# Perform the optimization to minimize variance
result = minimize(portfolio_variance, W_a, args=(C,), method='SLSQP',
                  bounds=bounds, constraints=constraints)

# Extract the optimal weights
mv_weights = result.x
```

**Maximum Sharpe Ratio Optimization:** in the code, the neg_sharpe_ratio() function calculates the portfolio weights that maximize the Sharpe ratio, a key measure of risk-adjusted return. The Sharpe ratio represents the excess return per unit of risk, considering the difference between the portfolio's return and a specified risk-free rate. The function takes as inputs the portfolio weights, expected returns, covariance matrix, and the risk-free rate, and returns the negative Sharpe ratio to be minimized (since optimization routines minimize by default).

This method is flexible because it allows for real-world constraints, such as:

1. No short-selling: Ensuring weights are non-negative (i.e., between 0 and 1).

2. Fully invested portfolio: Ensuring that the sum of the weights equals 1.

First, we perform portfolio optimization with the goal of maximizing the Sharpe ratio that adjusts the weights to maximize the Sharpe ratio by considering the risk (covariance) and the expected return (mu). The portfolio weights generated w_max_sharpe as 'initial_weights' from Max Sharpe Portfolio are passed to the quadratic optimization and the covariance matrix is passed as an argument to minimize the portfolio variance for optimized portfolio weights (see Table 14).

After determining the Max Sharpe Ratio Portfolio, we refine the portfolio by further minimizing risk (variance) to generate optimized weights. The quadratic optimization process uses Sequential Least Squares Programming (SLSQP), an iterative method that efficiently handles these constraints: no short-selling (weights between 0 and 1) and a fully invested portfolio (weights sum to 1).

Table 14: Max Sharpe Portfolio Initial Weights and Optimized Portfolio Weights using Constraints (no short selling)

Max Sharpe Portfolio:

| | ETF | Max Sharpe Weight |
|---|---|---|
| 0 | VUG | 0.188 |
| 1 | SPHD | 0.081 |
| 2 | MTUM | 0.084 |
| 3 | ACWI | 0.100 |
| 4 | XLK | 0.174 |
| 5 | XLE | 0.122 |
| 6 | IEO | 0.076 |
| 7 | GLD | 0.174 |

Optimized Portfolio Weights:

| | ETF | Optimized Weights |
|---|---|---|
| 0 | VUG | 0.000 |
| 1 | SPHD | 0.068 |
| 2 | MTUM | 0.115 |
| 3 | ACWI | 0.139 |
| 4 | XLK | 0.000 |
| 5 | XLE | 0.000 |
| 6 | IEO | 0.000 |
| 7 | GLD | 0.678 |

The result is a portfolio that maximizes the Sharpe ratio, making it ideal for investors looking to optimize their risk-adjusted returns in a real-world context, where constraints are important for practical portfolio construction. This optimization-based method is more robust than closed-form solutions, as it accounts for constraints typically required in portfolio management, making it suitable for real-world applications.

Max Sharpe Ratio Portfolio: Return = 2.91%, Volatility = 10.22%, Sharpe Ratio = -0.21

Max Sharpe Ratio Optimized Portfolio: Return = 0.81%, Volatility = 4.23%, Sharpe Ratio = -1.00

Table 15: Performance of the Max Sharpe Ratio Portfolios. The optimized portfolio (uses quadratic programming to minimize variance) has significantly reduced return while also reducing volatility, and a much lower Sharpe ratio.

Quadratic programming optimization is designed to minimize volatility but, by doing so, may direct too much capital to low-volatility assets that have little return potential. We could amend the code by introducing a minimum expected return constraint in the optimization to prevent the portfolio from becoming too conservative.

**Higher-order Moments:** in the code, we implemented higher-order moments such as minimizing coskewness and maximizing cokurtosis:

1. Coskewness: This measures the relationship between an asset's returns and the portfolio's returns. A portfolio with positive coskewness tends to have higher potential for positive extreme returns, and negative coskewness indicates higher downside risk.

2. Cokurtosis: This refers to the relationship between an asset's returns and the portfolio's return distribution in terms of "fat tails" or extreme events. Higher cokurtosis indicates a greater likelihood of extreme events.

In portfolio optimization, traditional mean-variance optimization does not account for these higher-order moments. We will modify the portfolio optimization problem by adding coskewness and cokurtosis to the objective function. This approach is suitable for investors who care not only about return and volatility but also about asymmetric risks (e.g., skewness) and tail risks (e.g., kurtosis).

Referencing this sample code[11], our implementation in project code is using the functions coskew() and cokurt() to compute the coskewness and cokurtosis matrices for the assets based on historical returns. These matrices are then used to adjust the portfolio optimization process.

The portfolio's expected return is relatively low at 0.81% as shown above. This low return suggests that the optimizer is prioritizing risk management (i.e., limiting downside risk) over maximizing returns. The large allocation to GLD likely reflects this emphasis on risk aversion.

[11] https://stackoverflow.com/questions/41890870/how-to-calculate-coskew-and-cokurtosis

Table 16: Weight Allocations and Performance of the Higher Order Moments Portfolio

```
Higher Order Moments Portfolio Return: 0.81%
Higher Order Moments Portfolio Volatility: 4.23%
Higher Order Moments Portfolio Sharpe Ratio: -1.0
Higher Order Moments Portfolio Coskewness: 0.89
Higher Order Moments Portfolio Cokurtosis: 9.8
```

```
Optimized Portfolio Weights:
    ETF  Optimized Weights
0   VUG              0.191
1  SPHD              0.277
2  MTUM              0.000
3  ACWI              0.000
4   XLK              0.000
5   XLE              0.000
6   IEO              0.000
7   GLD              0.532
```

The optimizer has significantly reduced risk through its allocation, particularly through the high weight in GLD. Sharpe ratio is negative and underperforming the risk-free rate. Favorable positive coskewness indicates that the portfolio is more likely to experience positive outliers. High cokurtosis indicates that the portfolio is exposed to fat tails or extreme events.

**Minimize Tracking Error:** Tracking error is the standard deviation of the difference between the portfolio returns and the benchmark returns. To minimize tracking error, we optimize the portfolio to minimize the deviation between the portfolio's returns and the returns of SPY benchmark. In the code implementation, we use a manual_optimization() function to adjust the portfolio weights by considering both the minimum return constraint and excess returns over the benchmark. If the initial weights do not satisfy the minimum return constraint (i.e., 2%), it adjusts the weights to focus on assets that have higher excess returns.

Table 17: Weight Allocations and Performance of the Min Tracking Error Portfolio

```
Min Tracking Error Portfolio Return: 1.72%
Min Tracking Error Portfolio Volatility: 6.86%
Min Tracking Error Portfolio Sharpe Ratio: -0.48
Min Tracking Error Value: 7.29%
```

```
Check on length alignment: False
Optimized Portfolio Weights:
       ETF  Optimized Weights
ETF
ACWI  ACWI              0.037
GLD    GLD              0.054
IEO    IEO              0.282
MTUM  MTUM              0.029
SPHD  SPHD              0.000
VUG    VUG              0.210
XLE    XLE              0.140
XLK    XLK              0.247
```

The portfolio gives significant weight to IEO (28.2%), XLK (24.7%), and XLE (14.0%), which are tied to oil, gas, and technology. Exclusion of SPHD (high dividend stocks) suggests that these stocks are not seen as contributing to the tracking error minimization. The portfolio allocates 21% to VUG (large growth stocks) and 5.4% to GLD (gold). This balances the portfolio by providing exposure to growth while maintaining a minor defensive position with gold as a hedge against market volatility. Overall, the return at 1.72% is poor performance, relatively low volatility and a negative Sharpe ratio indicates that the portfolio is underperforming against the risk-free rate.

Table 18 : All Portfolio Optimizations Performance Metrics

Portfolio Metrics:

| | Metric | Mean-Variance Portfolio | Max Sharpe Portfolio | Higher-Order Moments Portfolio | Tracking Error Minimized Portfolio |
|---|---|---|---|---|---|
| **0** | Expected Return | 0.81% | 2.91% | 0.81% | 0.06% |
| **1** | Standard Deviation (Risk) | 4.23% | 10.22% | 4.23% | 1.37% |
| **2** | Sharpe Ratio | -1.00 | -0.21 | -1.00 | -3.62 |
| **3** | Tracking Error | N/A | N/A | N/A | 0.14% |

The portfolio metrics reveal contrasting performances across different optimization strategies. The Max Sharpe Portfolio achieves the highest expected return at 2.91%, though with increased risk at 10.22%, resulting in a relatively low Sharpe Ratio of -0.21.

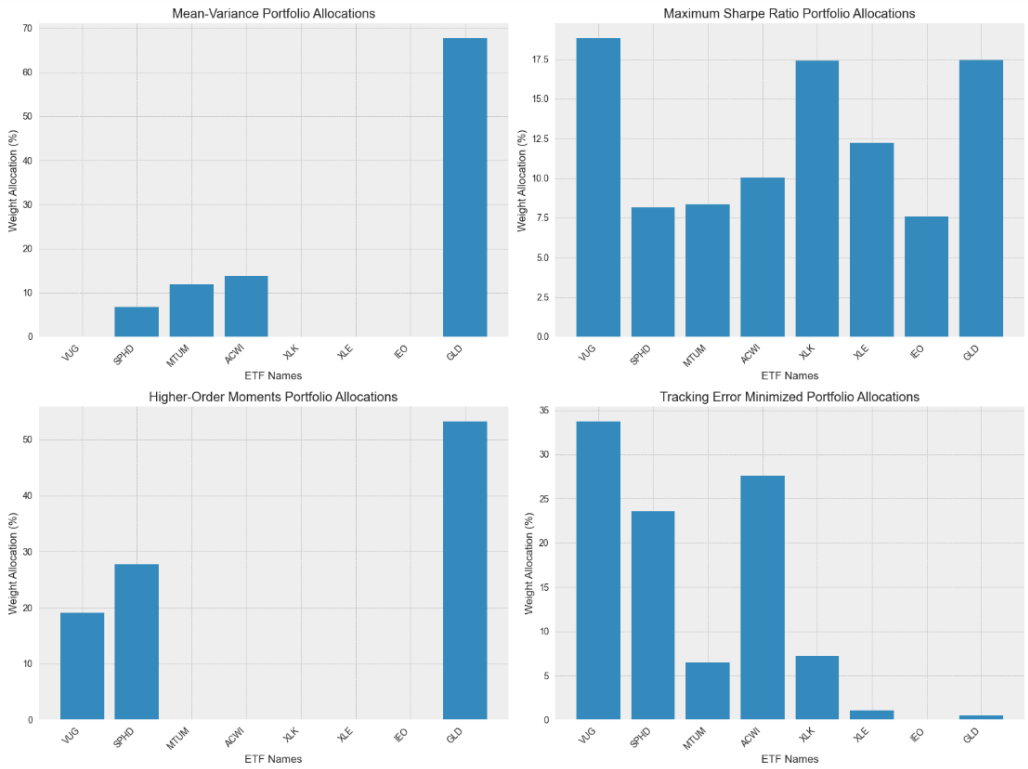



Figure 11: Portfolio allocations for different portfolio optimizations

The Mean-Variance and Higher-Order Moments Portfolios both deliver modest 0.81% returns with lower risk (4.23%), but their Sharpe Ratios are notably negative at -1.00, indicating underperformance relative to risk. The Tracking Error Minimized Portfolio achieves the lowest risk at 1.37%, but it is extremely low 0.06% return and -3.62 Sharpe Ratio reflect very conservative performance, despite minimizing deviations from the benchmark with a tracking error of 0.14%.

### 4.10 Evaluation of Risk Aversion through the Lenses of Three Distinct Investor Profiles

Evaluating risk aversion through the lenses of three distinct investor profiles—Trustee, Market, and Kelly Investor—while aiming to maximize the log of wealth involves understanding the risk tolerance and investment objectives of each investor type.

1. Trustee Investor:

Profile Characteristics: Typically conservative, prioritizing capital preservation over aggressive growth. Risk aversion is high.

Implementation Strategy: For a Trustee investor, maximizing the log of wealth involves a portfolio with lower volatility and a focus on stable, income-generating assets. This could mean a higher allocation to bonds, dividend-paying stocks, or ETFs with a history of lower volatility. The optimization process should incorporate constraints that minimize risk, perhaps using a modified utility function that heavily penalizes portfolio variance.

2. Market Investor:

Profile Characteristics: Represents the average market participant with a moderate risk tolerance. This investor seeks a balance between risk and return, mirroring the broader market performance.

Implementation Strategy: For the Market investor, the portfolio should aim to replicate the risk-return profile of the

market benchmark or index. Maximizing the log of wealth involves aligning the portfolio's risk level with that of the market, using a mix of stocks and bonds that closely approximates the market's composition. The optimization can be tuned to achieve a portfolio beta close to 1, indicating that the portfolio's movements are expected to closely follow those of the market.

3. Kelly Investor:

Profile Characteristics: Aggressive and focused on maximizing long-term growth. This investor is willing to tolerate higher short-term volatility for the potential of higher returns. The Kelly criterion, which maximizes the expected logarithm of wealth, naturally aligns with this investor's objectives.

Implementation Strategy: For the Kelly investor, the portfolio optimization focuses on maximizing the expected logarithmic growth rate of the portfolio. This involves calculating the optimal fraction of the portfolio to allocate to each investment to maximize the expected log return, considering the probability of winning (gains) and losing (losses) on each bet or investment. The Kelly formula or its adaptations can be applied directly, adjusting the leverage to suit the investor's appetite for risk (see detailed implementation in Section 4.11). This strategy often leads to a more concentrated portfolio with higher expected growth and volatility.

In Project Code implementation, we adjust portfolio allocations based on different risk tolerance levels, allowing for more tailored investment strategies. The risk levels are defined as Trustee (more conservative, with a risk aversion of 3.07), Market (a standard risk level of 1), and Kelly (aggressive, with a lower risk aversion of 0.5).

Starting with the optimized weights from four different portfolio strategies - Mean-Variance (MV), Max Sharpe Ratio, Tracking Error Minimized, and Higher-Order Moments (HOM) - the code adjusts these weights based on the chosen risk aversion level. This adjustment is done by dividing the initial weights by the risk aversion level, which means portfolios for more risk-averse investors (Trustee) will have smaller allocations, while risk-tolerant investors (Kelly) will have larger ones.

The final output is in allocation_df that compares the adjusted weights for each portfolio strategy and risk level, making it easy to see how the allocations change depending on the investor's risk tolerance. This method is useful for portfolio managers who want to tailor their strategies to different client profiles without completely changing the underlying optimization method.

Table 19: Adjusted portfolio weights for different ETFs across four portfolio optimization strategies—Mean-Variance, Max Sharpe Ratio, Tracking Error Minimization, and Higher-Order Moments—under three risk aversion levels: Trustee (conservative), Market (standard), and Kelly (aggressive)

| | ETF | MV Trustee | MV Market | MV Kelly | Max Sharpe Trustee | Max Sharpe Market | Max Sharpe Kelly | Tracking Error Trustee | Tracking Error Market | Tracking Error Kelly | Higher-Order Moments Trustee | Higher-Order Moments Market | Higher-Order Moments Kelly |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | VUG | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.110 | 0.337 | 0.675 | 0.062 | 0.191 | 0.382 |
| 1 | SPHD | 0.022 | 0.067 | 0.135 | 0.022 | 0.068 | 0.136 | 0.077 | 0.236 | 0.472 | 0.090 | 0.277 | 0.553 |
| 2 | MTUM | 0.038 | 0.118 | 0.236 | 0.038 | 0.115 | 0.230 | 0.021 | 0.065 | 0.129 | 0.000 | 0.000 | 0.000 |
| 3 | ACWI | 0.045 | 0.138 | 0.276 | 0.045 | 0.139 | 0.277 | 0.090 | 0.276 | 0.551 | 0.000 | 0.000 | 0.000 |
| 4 | XLK | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.023 | 0.072 | 0.144 | 0.000 | 0.000 | 0.000 |
| 5 | XLE | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.003 | 0.010 | 0.021 | 0.000 | 0.000 | 0.000 |
| 6 | IEO | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 7 | GLD | 0.220 | 0.676 | 1.353 | 0.221 | 0.678 | 1.357 | 0.001 | 0.005 | 0.009 | 0.173 | 0.532 | 1.064 |

The portfolio allocation shifts significantly as the risk aversion level moves from Trustee to Kelly. More aggressive allocations (Kelly) favor growth-oriented and high-risk assets, while conservative allocations (Trustee) focus on stability and diversification. The Higher-Order Moments strategy leans towards defensive assets like GLD, while Tracking Error Minimization focuses on balanced exposures to global equities like ACWI.

### 4.11 Kelly Criterion Portfolio Strategy

In practical application of the Kelly strategy across different stocks or ETFs, investors often draw inspiration from successful practitioners such as Warren Buffett, Charlie Munger, and Mohnish Pabrai, who utilize the Kelly Criterion as a guide to determine the size of their investments. The Kelly Criterion is a mathematical formula that helps allocate capital optimally based on the perceived edge in a given profitable investment opportunity.

For instance, in a simple game scenario of a dollar, the Kelly Percentage is calculated as follows:

$$K = \text{Probability of Winning} - \frac{(1 - \text{Probability of Winning})}{\text{Winnings as a Percent}}$$

Suppose the calculated Kelly Percentage is 0.167 or 17% of the available balance. This means that for the first bet in the game, it would be wise to wager 17 cents. The beauty of the Kelly Criterion lies in its ability to provide a systematic approach that helps prevent significant capital loss, allowing investors to stay in the game even after a loss.

Importantly, the Kelly Formula is most effective when applied to situations with a profitable opportunity, where the odds are in favor, and potential gains outweigh potential losses. This aligns with the principles of the stock market and various investments.

To illustrate this in the context of the stock market, let us consider a stock with a win probability of 45% and a loss probability of 55%, with the potential gain being a conservative estimate of a 150% increase in stock price, the investor applies the Kelly Formula:

$$K = 0.45 - \frac{(1 - 0.45)}{1.5}$$

For this example, the calculated Kelly Percentage is 0.083 or 8%. This indicates the proportion of the investor's available capital that would be judiciously allocated to this particular investment opportunity, considering the perceived risks and rewards. In our code implementation:

- Instead of exact "win/loss" probabilities, we use expected returns based on market views or historical data.
- Instead of the "payoff multiple," we consider the risk/reward characteristics, such as volatility or Sharpe ratios, to estimate the edge in investing.

Referencing Yves Hilpisch's Python for Algorithmic Trading book by O'Reilly, we adapted the Kelly logic for portfolio construction by considering the expected returns and the covariance (risk) of those returns. It calculates an optimal allocation based on the risk-reward profile of each asset.

The kelly_allocation() function in the Project Code computes the optimal weights based on the Kelly criterion using:

1. Expected returns (returns.mean()).
2. Covariance matrix (returns.cov()), representing the variability of asset returns.
3. Risk-free rate (annual_rfr), which is typically set to 3%.
4. The weights are then normalized using an L1 norm to ensure they sum up to 1, representing a fully invested portfolio.

The formula used to calculate the Kelly weights is:

$$W = inv(Cov) \times \left(\mathrm{E[R]} - \mathrm{R}_f\right)$$

Where:

- W is the Kelly optimal weights.
- Cov is the covariance matrix of asset returns.
- E[R] is the expected return vector.
- $R_f$ is the risk-free rate.

The kelly_strategy function simulates portfolio performance over time using the calculated Kelly weights. The portfolio return is computed as a weighted sum of individual asset returns at each time step. Three Kelly fractions are used: 0.5 (half Kelly), 0.66 (two-thirds Kelly), and 1.0 (full Kelly), each representing different levels of risk aversion, and the key steps are:

1. Equity and Capital Initialization: The equity starts at 1, and the capital is initialized by multiplying equity by the Kelly fraction (e.g., 0.5, 0.66, or 1.0).

2. Portfolio Return Calculation: The portfolio return is calculated as the weighted sum of asset returns using the Kelly weights.

3. Capital and Equity Updates: The capital and equity are updated iteratively based on the portfolio's performance over time.

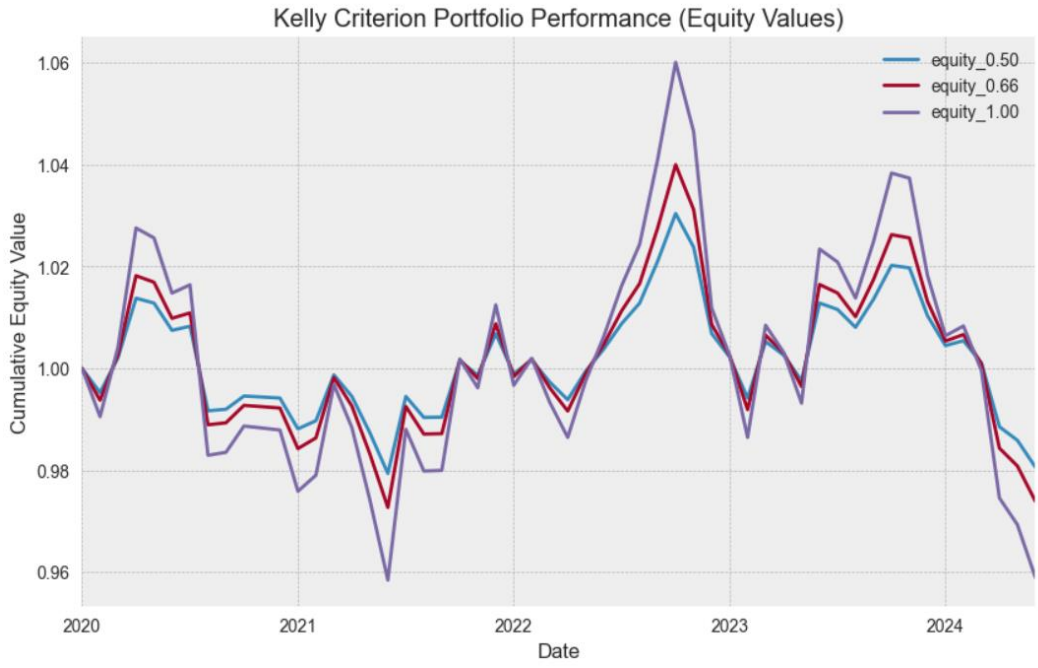



Figure 12: The cumulative returns and equity values for each Kelly fraction (0.5, 0.66, 1.0) are plotted over time, allowing a visual comparison of portfolio performance under varying risk levels. The plot shows how higher fractions of Kelly (full Kelly) lead to higher cumulative returns but also higher risk (as evidenced by greater volatility), while lower fractions (half Kelly) offer more stability but lower returns.

Referring to Figure 12 above and Figure 13 in the next page, the scaling of the Kelly allocation (fractional vs full) had little impact on the portfolios' performance, and the portfolios were conservative and did not capture much upside in the market. The conservative returns could be due to the 'returns.mean()' being low and used in the Kelly allocation calculation, and normalization of the weights using the L1 norm led to conservative weights as it reduces exposure to more volatile assets.

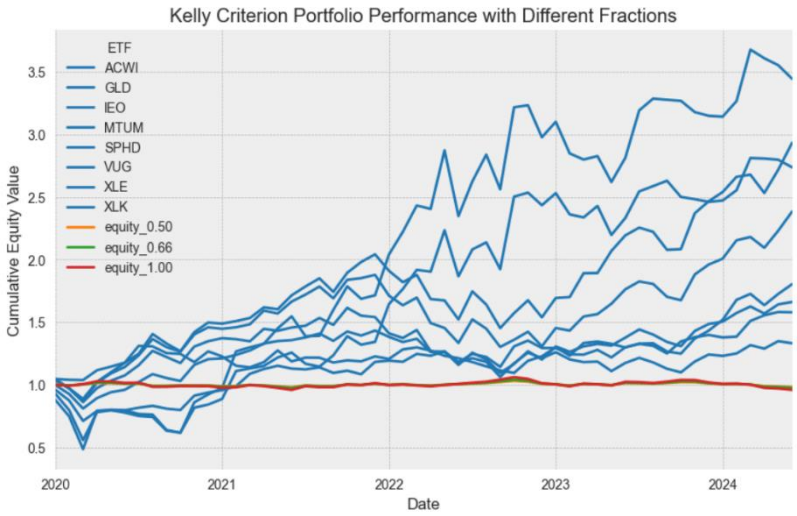



Figure 13: The plot on performance of Kelly portfolios vs. ETFs where the three Kelly allocations (0.5, 0.66, and 1.0) exhibit almost flat performance over time, with very limited growth compared to the ETFs, which display much higher cumulative returns.

In conclusion, while no investing system is infallible, the Kelly formula serves as a valuable tool to allocate funds effectively to opportunities with favorable risk-reward profiles. It empowers investors to be more decisive, particularly when faced with high-reward opportunities where the odds are in their favor. The key is to apply the formula judiciously, considering the specific characteristics of each investment and the overall portfolio strategy.

# 5 Future Work

## 5.1 Risk-Sensitive Asset Management Model

In future work, we can extend the traditional portfolio optimization by incorporating the investor's risk preferences directly into the decision-making process using the risk-sensitive asset management model. Unlike the classical mean-variance optimization, which only considers the trade-off between return and volatility, a risk-sensitive model accounts for higher moments of the distribution or specific investor risk aversion characteristics.

The risk-sensitive asset management model's optimal asset allocation is:

$$h^{\theta}(t) = \frac{1}{\theta + 1}(\sum \sum')^{-1}\left[\tilde{a} + \tilde{A}\,X\,(t) - \theta \sum \wedge'\,(Q(t)\,X(t) + q(t))\right]$$

The optimal strategy for fund separation is constituted of positions in two funds:

1. An allocation of a fraction $\frac{1}{\theta+1}$ of wealth to the Kelly portfolio:

$$h^{K}(t) = (\sum \sum')^{-1}\left[\tilde{a} + \tilde{A}\,X\,(t)\right]$$

2. An allocation of a fraction $\frac{\theta}{\theta+1}$ of wealth to the 'intertemporal hedging portfolio' (non-cash as cash has risk):

$$h'(t) = (\sum \sum')^{-1}\left[\sum \wedge' \left(Q(t)\,X(t) + q(t)\right)\right]$$

$(\sum \sum')^{-1}[\sum \wedge']$ is a term projection of the factor risk on to the subspace of asset risk. Objective is to use the assets to hedge the risk coming from the factors. Similar to Ordinary Least Squares estimation of non-tradable factor risk using

tradable asset risk. The terms used in the formulas above are:

- Θ: investor's risk aversion level
- $\tilde{a}$ & $\tilde{A}$: dynamics of asset returns
- $Q(t)$ & $q(t)$ : external economic factors
- $(\sum \sum')^{-1}$ : inversion of the covariance matrix of asset returns, used for optimizing the portfolio against the perceived risks.
- $\tilde{a} + \tilde{A}\, X\,(t)$: the expected returns on assets, adjusted for their exposure to underlying factors X(t).
- $\theta \sum \wedge' (Q(t)\, X(t) + q(t))$: accounts for the adjustment in allocation due to the intertemporal hedging demand, related to external economic factors impacting future investment opportunities.

We can incorporate risk preferences into the portfolio optimization so as to implement the risk-sensitive asset management model:

- Investor's risk aversion level (Θ): we determine the risk aversion parameter that best represents our target trading strategy's risk profile. This parameter will be crucial in adjusting the allocation between the Kelly portfolio and the intertemporal hedging portfolio.
- Optimal asset allocation model: Adapt the risk-sensitive asset allocation model to calculate the optimal weights for the ETFs in the portfolio by estimating the expected returns ($\tilde{a} + \tilde{A}\, X\,(t)$) and the covariances ($(\sum \sum')^{-1}$) of the ETFs, as well as identifying the relevant economic factors ($Q(t)$ & $q(t)$) that may affect their future performance. We will use discounted asset prices rather than nominal asset prices in the portfolios section because the optimal investment policy depends on the excess return over the short-term rate (risk premia) and not on the nominal rates of return.
- Black-Litterman Model: Using the Black-Litterman model for model inputs to generate expected returns for the ETFs, incorporating investor's views and the market equilibrium. This can serve as the $\tilde{a} + \tilde{A}\, X\,(t)$ term in the risk-sensitive model.
- Fama-French Factors: Using the five factor Fama-French model to identify the underlying economic factors that affect the returns of the ETFs. These factors can be part of the $Q(t)\, X(t) + q(t)$ term to forecast changes in investment opportunities.

By evaluating investor risk preferences through different profiles—such as Trustee, Market, and Kelly Investor—we aim to maximize the logarithmic utility of wealth. Risk management tools, such as Value at Risk (VaR) and Conditional Value at Risk (CVaR), will be used to assess the overall risk of the optimized portfolios. Dynamic hedging strategies are employed to manage risk exposures effectively. The optimized portfolios will be regularly monitored and rebalanced based on market conditions and updated forecasts. Performance is evaluated against benchmarks, with ongoing adjustments to enhance returns and mitigate risks in real time.

### 5.2 LSTM Model to Predict Future Expected Returns

In future work, we propose training an LSTM model on historical ETF datasets to predict future returns, with these predictions treated as 'expert views.' These machine learning-informed predictions may offer a more nuanced, data-driven alternative to traditional methods, potentially diverging from or refining the market consensus. By integrating these expert views into the portfolio optimization process, we aim to complement the market equilibrium with predictive insights.

To dynamically assess unobserved risk factors, a Kalman filter could be incorporated, leveraging LSTM forecasts as expert observations. This approach would account for the inherent uncertainty in forward-looking predictions. The Kalman filter would dynamically update risk factor estimates, adjusting them as new market data and LSTM predictions become available. This integration could enhance the portfolio's adaptability to real-time market shifts by optimizing portfolio allocations based on continuous updates to the risk factors.

The future development of this LSTM-Kalman approach would focus on optimizing the portfolio for maximum expected logarithmic utility, ensuring that strategies remain adaptive to changing market dynamics while staying within traditional constraints such as short selling prohibitions, leverage limits, and diversification requirements.

### 5.3 Devise Black-Litterman Views using Deep Learning

To further enhance our portfolio optimization process, future work will focus on incorporating deep learning techniques, particularly LSTM networks, into the generation of Black-Litterman views. The limitations of traditional mean-variance (MV) optimization, such as the sensitivity to input parameters and resulting corner solutions, can be mitigated by using LSTM models to provide more robust and balanced assessments of risk and return:

1. Corner Solution Issue: Traditional mean-variance (MV) optimization sometimes results in extreme allocations, often investing heavily in assets with perceived high returns but significant risk. By using LSTM models to generate views on expected returns, the risk of corner solutions is reduced, offering a more nuanced and balanced assessment of risk and return. LSTM models incorporate broader market dynamics and data patterns, providing improved predictive performance.

2. Heteroskedasticity and Variance Estimation: Variance estimation, especially when affected by heteroskedasticity, can undermine the stability of portfolio optimizations. LSTMs can analyze temporal patterns in volatility to better predict future variances. This process can be further enhanced by applying denoising and shrinkage techniques:

    a. Denoising: LSTM models can filter out noise, producing more reliable variance estimates. Coupled with shrinkage, estimates can be adjusted towards long-term averages for added stability.

    b. Black-Litterman (BL) Views: LSTM models can also incorporate external data sources, such as macroeconomic indicators and news sentiment, to generate forward-looking views. These views feed into the Black-Litterman model, offering a dynamic approach to adjusting portfolio weights based on forecasted market conditions.

3. Risk Budgeting and Risk Parity: LSTMs also assist in risk budgeting by forecasting the risk contributions of individual assets. This allows for the construction of risk parity portfolios, where assets are weighted according to their risk contributions rather than solely expected returns. By accurately predicting both return and risk characteristics, LSTMs improve portfolio diversification and stability.

The future integration of LSTM networks into portfolio construction would directly address some of the core challenges of traditional MV optimization, enhancing predictive accuracy for returns and volatility. Additionally, the deep learning-driven Black-Litterman views would provide a more dynamic, forward-looking approach to adjusting portfolio weights based on both macroeconomic data and sentiment analysis.

Performance Evaluation and Next Steps: the future work will focus on testing LSTM-based Black-Litterman models through backtesting platforms like Backtrader, comparing the model's performance against benchmarks like the Dow Jones Industrial Average (DJIA) and traditional Markowitz portfolios. Preliminary studies12 indicate that LSTM-enhanced models can outperform traditional approaches in both risk-adjusted and absolute returns by improving portfolio stability and reducing extreme allocation shifts.

[12] Research Review: Novel Asset Allocation: LSTM-Based Black-Litterman Model, July 24, 2020, by Peter Lin

Additionally, incorporating techniques like stock network centrality and other machine learning-driven insights could further enhance short-term and intraday trading performance. The development of this LSTM-based approach will offer an advanced, adaptive framework for portfolio construction that integrates deep learning with traditional models, improving overall portfolio robustness and responsiveness to market changes.

# 6 Conclusions

In this project, we developed a sophisticated portfolio construction framework by integrating traditional and modern methodologies, including the Fama-French Five-Factor model and the Black-Litterman model. Our goal was to create a diversified portfolio that adapts to changing market conditions while reflecting investor views through a blend of systematic backtesting, regression-based techniques, and optimization strategies.

We calculated excess returns relative to the global market proxy, ACWI, and applied the Black-Litterman model to adjust implied equilibrium returns based on subjective investor views. The use of investor views allowed us to refine the market consensus, incorporating unique insights derived from predictive models. Portfolio optimization was further enhanced through the combination of techniques such as Mean-Variance Optimization, Maximum Sharpe Ratio, and Minimum Tracking Error, providing a comprehensive allocation strategy across three distinct risk profiles: Trustee, Market, and Kelly Investor. These techniques enabled us to balance risk and return while maintaining a disciplined approach to diversification.

A key contribution of our work was the application of factor loadings to optimize portfolio exposure to key risk factors—market, size, value, profitability, and investment—using the Black-Litterman framework. Additionally, we developed a visual evaluation approach for multi-factor models, offering an alternative to traditional mean-variance optimization methods. The inclusion of the Kelly Criterion further enhanced long-term wealth growth by focusing on assets with strong potential and favorable risk-return profiles.

While our approach successfully constructed a robust, well-diversified portfolio, some limitations remain. Expanding the model to account for additional sentiment-driven data could offer further insights. Additionally, refining the creation of Black-Litterman views using deep learning or using LSTM to predict future expected returns so as to better capture meaningful states of the data could potentially enhance the model's portfolio performance. Future research could address these areas to further improve portfolio performance across different market environments.

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